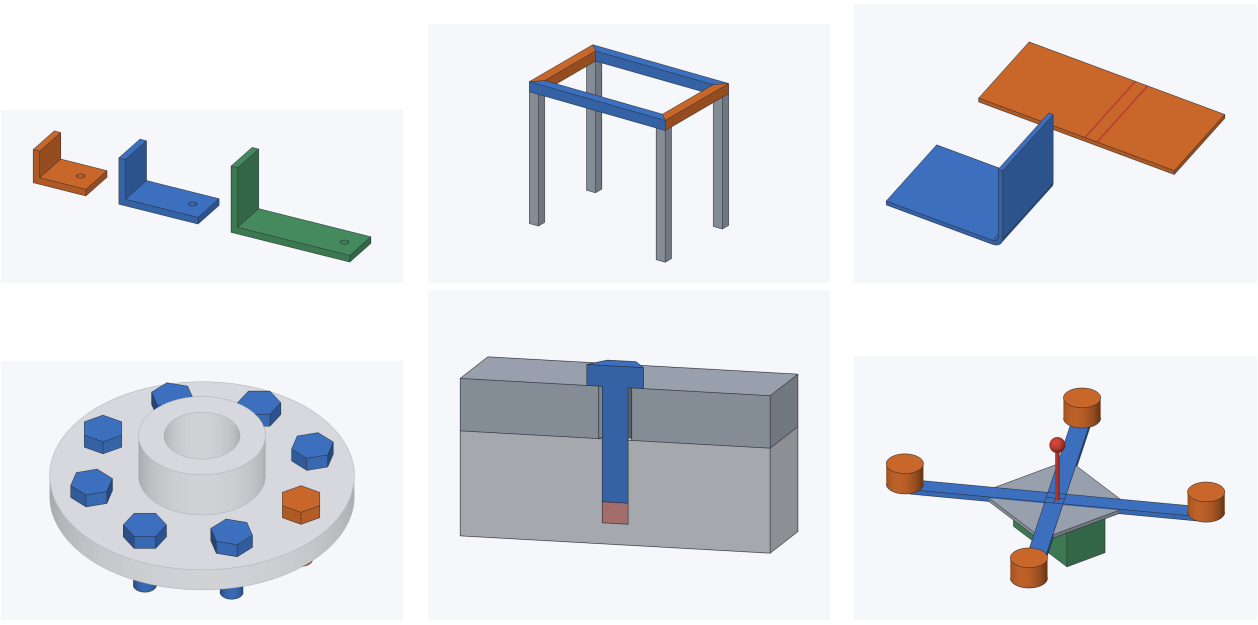
BruceWorx Roadmap

Full Fusion 360 / SolidWorks parity, from first sketch to released drawing

Epics, design rationale and theory

Status	Draft for review
Date	19 October 2026
Base	`main` @ `696260b`, FreeCAD 1.1.4 (BruceWorx build)
Scope	12 tracks, 81 epics, milestones M0–M6 (`roadmap-epics.csv`)



All six models were built and recomputed in headless FreeCAD 1.1.4 by the scripts in `scripts/`. The renders are working figures; they will be regenerated after the UI redesign (`UIX-09`).

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1 Summary

Goal. BruceWorx (FreeCAD) reaches *full parity with Fusion 360 and SolidWorks* across the whole product lifecycle, so that it can become the CAD system of record for BruceWorx products. That covers sketching and part modeling, assembly, simulation, ASME drawings, part and drawing management, and the files sent to the 3D printer, the CNC shop and the laser cutter. This replaces the earlier position (FC-T01) that FreeCAD only complements Fusion. The switch itself is the last milestone (M6) and is gated on the parity matrix in Appendix A.

How this document was built. The previous version of this file was a UI plan (Fusion-style layout, patterns, command line). It is now one chapter's worth of material inside a roadmap with **12 tracks and 81 epics**. The inputs were:

- a five-round interview with you (18–19 Oct 2026), recorded in §1.1;
- the BruceWorx Ops "work" list (191 items, marked *sample data*), used as *ideas only.* Appendix B maps every idea group to the epic that absorbs it;
- a capability survey of FreeCAD 1.1.4 in this repository (grep of the actual sources and headless experiments), compared with Fusion 360 and SolidWorks 2026 public material;
- the repository's merged work (`[BW]` commits #3–#9), which marks four epics done and five in progress.

Granularity. The roadmap stops at *epics.* Each epic has a scope, dependencies, an acceptance test and the ideas it absorbs. Tasks are broken down when an epic starts. `roadmap-epics.csv` carries the same 81 epics for import into Ops. It is generated from `scripts/epics.py`, together with the tables in this document, so the two cannot drift.

Styling comes later. The UI and styling track runs after M2. The figures here are working renders and will be regenerated for the new UI (`UIX-09`).

1.1 Decisions from the interview

Topic	Decision
End state	Full Fusion 360 / SolidWorks parity; FreeCAD becomes system of record at M6
Ops task list	Ideas only; fresh IDs (`BW-`*`TRK`*`-nn`), origins kept in Appendix B
Manufacturing	FDM printing (PETG/ABS), CNC machining, sheet metal / laser. Not injection molding
First stage	Part and assembly modeling (main track)
Track order	Modeling/assembly first; **PDM evaluation and drawings run in parallel from the start**
Core modeling	Configurations; direct edit on imported STEP; parameters and top-down; threads and hardware
Advanced modeling	Freeform / T-spline-like; surfacing parity; mesh → solid; frames / weldments
File model	SolidWorks-style: **one file per part**, each with its own part number; assemblies link to files
Assembly extras	In-context editing; large-assembly mode; smart fastener mating; dynamic motion study
Simulation	Static + contact, modal and buckling, thermal and thermal stress. CFD later
FDM materials	Isotropic + knockdown by default; orthotropic by print orientation available per study
Drawing standard	ASME Y14.5, third-angle projection
Drawings	Associative BOM + balloons; GD&T and datums with model-driven callouts; revision table and release; auto-generate drawing
PDM home	**Evaluate first** (Ops as PLM vs Ops + Git LFS vs in-FreeCAD vault vs open-source PLM)
Lifecycle	WIP → In Review → Released → Obsolete. Numeric pre-release revisions (01, 02, . . .), letter at first release (A), then ECR → ECO for every later letter (ASME Y14.35 + SW-PDM practice)

Topic	Decision
Revision vs P/N	Revision is metadata, separate from the part number; drawings and packages print `P/N-REV`
Team	2–5 concurrent editors: check-out locks, review and approval
Outputs	3D-print prep; CNC CAM; sheet-metal flat DXF; release package
Kept idea areas	AI / MCP assistant; KiCad bridge; robotics and drone kits; kernel study
UI items	Patterns → Modeling; workspaces, panels, toolbar → UI track (later); command line → Automation
Upstream	**Track FreeCAD main** continuously (replaces "stay on 1.1.4, rebase at 26.3")
Deliverable	This document plus `roadmap-epics.csv`; epics only
Rule (standing)	Every feature is on by default, with an off switch on Preferences → BruceWorx

1.2 The lifecycle and where parity stands

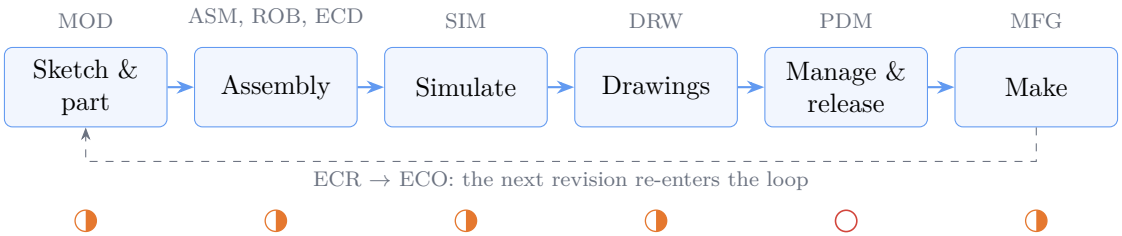


Figure 1: Lifecycle stages, the tracks that serve them, and FreeCAD 1.1.4's overall standing against Fusion/SolidWorks (● full, ◑ partial, ○ none). Part and drawing management is the only stage with nothing in core FreeCAD. Foundation (FND), automation (AUT), UI (UIX) and kernel (KRN) cut across all stages.

The detailed matrix (Appendix A, 71 capabilities) shows the pattern behind Figure 1:

- **Strong already:** parametric solids, joints and kinematic simulation (Assembly in 1.1), CalculiX/Elmer analysis types, drawing views, CAM and post-processors, 3MF and DXF export, Python.
- **Partial:** patterns (with a bug, §4.1), parameters, direct edit (Defeaturing only), surfacing, threads (drawing-side only), BOM (four automatic columns), GD&T (symbol library, not semantic), mesh → solid (faceted).
- **Missing:** configurations, freeform, weldments, interference, smart fasteners, dynamic motion, safety-factor and orthotropic results, auto drawings, *all* of PDM, slicer handoff, rendering, generative design.

1.3 Milestones

Milestones are *gates*, not calendar dates. Tracks run in parallel. A milestone closes when all of its epics pass acceptance on real BruceWorx parts. The Leonardo pedestal assembly `LE-02001` is the pilot throughout.

	Milestone	Exit criterion
M0	Foundation green	CI green on Linux and macOS arm64; tagged release; nightly rebase on upstream main; review fixes in
M1	Model it	LE-02001 modelled in FreeCAD from scratch: one file per part, patterns, configurations, threads, joints, interference-free
M2	Document & release it	LE-02001 rev A released *from FreeCAD*: ASME drawings, BOM, PDM lifecycle, release package
M3	Make it	Printed, machined and laser-cut parts made from FreeCAD outputs without touching Fusion
M4	Validate it	Static, modal, buckling and thermal studies on LE parts, verified against benchmarks
M5	Advanced	Surfacing, freeform, mesh → solid, in-context, large-assembly, dynamic motion, command line, assistant
M6	Switch system of record	Parity matrix met for the LE scope; Fusion data migrated; FC-T01 criteria satisfied
UI	UI and styling pass	After M2: workspaces, panels, fresh visual design, regenerated figures
BG	Background	Kernel study; runs at low priority throughout

Track	M0	M1	M2	M3	M4	M5	M6	UI	BG
Foundation & upstream	01 02 03 04 05 06 07								
Part modeling		01 02 03 04 05 11 12		06 07		08 09 10			
Assembly		01 02 03 04 05 06				07 08 09 10			
Drawings (ASME)			01 02 03 04 05 06 08	07					
Part & drawing management			01 02 03 04 05 06 07				08		
Simulation					01 02 03 04 05	06			
Manufacturing outputs			04	01 02 03					
Robotics & drone kits		01 03		02 04		05			
ECAD bridge (KiCad)		01		02 03 04					
AI & automation	01	02 03				04 05			
UI & styling								01 02 03 04 05 06 07 08 09	
Kernel study									01 02 03

Figure 2: Epics by track and milestone (number = epic, e.g. Part modeling `03` = `BW-MOD-03`). Colour shows status: **done**, **in progress**, to do. Generated from `scripts/epics.py`.

1.4 Critical path

1.5 All epics

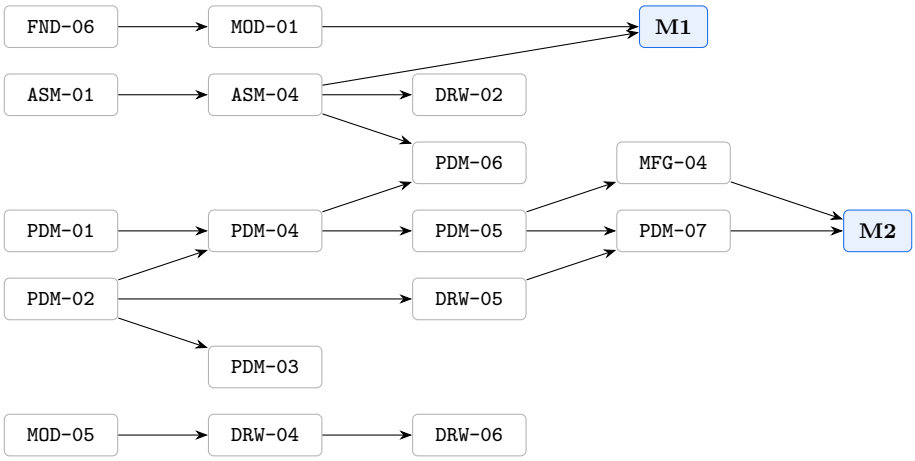


Figure 3: The dependencies that decide when FreeCAD can release a documented assembly (M2). The PDM evaluation (`PDM-01`) and data model (`PDM-02`) have no prerequisites, which is why they start on day one alongside modeling.

ID	Epic	Mile.	Depends on	Status
Foundation & upstream				
FND-01	Build & CI on Linux and macOS arm64	M0	—	In progress
FND-02	Packaging, branding & releases	M0	FND-01	In progress
FND-03	Track upstream FreeCAD main	M0	FND-01	To do
FND-04	Licences & notices	M0	—	To do
FND-05	Acceptance corpus & test harness	M0	—	To do
FND-06	Fixes found in review	M0	—	In progress
FND-07	On by default, each with an off switch	M0	—	Done
Part modeling				
MOD-01	Rectangular & circular patterns	M1	FND-06	In progress
MOD-02	Parameters & top-down design	M1	—	To do
MOD-03	Configurations	M1	MOD-02	To do
MOD-04	Direct edit on imported geometry	M1	—	To do
MOD-05	Threads, holes & hardware library	M1	—	To do
MOD-06	Sheet metal	M3	—	To do
MOD-07	Frames & weldments	M3	—	To do
MOD-08	Surfacing parity	M5	—	To do
MOD-09	Freeform (subdivision / T-spline-like)	M5	MOD-08	To do
MOD-10	Mesh to solid	M5	—	To do
MOD-11	Inspect tools	M1	—	To do
MOD-12	Sketcher parity audit	M1	—	To do
Assembly				
ASM-01	File model: one part per file	M1	—	To do
ASM-02	Assembly gap audit	M1	—	Done
ASM-03	Joint UX & limits	M1	ASM-02	In progress
ASM-04	Exploded views & BOM	M1	ASM-01	In progress
ASM-05	Sub-assemblies	M1	ASM-01	To do
ASM-06	Interference & clearance	M1	—	To do
ASM-07	In-context editing	M5	ASM-01 MOD-02	To do
ASM-08	Large-assembly mode	M5	ASM-01	To do
ASM-09	Smart fastener mating	M5	MOD-05 ASM-03	To do
ASM-10	Dynamic motion study	M5	ASM-03	To do
Drawings (ASME)				
DRW-01	ASME templates & title block	M2	—	To do
DRW-02	Associative BOM table & balloons	M2	ASM-04	To do

ID	Epic	Mile.	Depends on	Status
DRW-03	GD&T & datums (ASME Y14.5)	M2	—	To do
DRW-04	Hole & thread callouts from the model	M2	MOD-05	To do
DRW-05	Revision table & release stamp	M2	PDM-02	To do
DRW-06	Auto-generate drawing	M2	DRW-01 DRW-04	To do
DRW-07	Sheet-metal flat & cut-list drawings	M3	MOD-06 MOD-07	To do
DRW-08	Publish PDF/DXF	M2	DRW-01	To do
Part & drawing management				
PDM-01	PDM evaluation & decision	M2	—	Done
PDM-02	Data model & lifecycle spec	M2	—	In progress
PDM-03	Numbering service	M2	PDM-02	In progress
PDM-04	FreeCAD vault client	M2	PDM-01 PDM-02	To do
PDM-05	Lifecycle & change control	M2	PDM-04	To do
PDM-06	BOM sync & where-used	M2	ASM-04 PDM-04	To do
PDM-07	Pilot: LE-02001 & CAD register	M2	PDM-05 DRW-05	To do
PDM-08	Migration from Fusion	M6	PDM-07	To do
Simulation				
SIM-01	Study setup (Fusion-style)	M4	—	To do
SIM-02	Contacts, bolts & preload	M4	SIM-01	To do
SIM-03	Materials incl. FDM	M4	SIM-01	To do
SIM-04	Results & reports	M4	SIM-01	To do
SIM-05	Verification benchmarks	M4	SIM-01	To do
SIM-06	CFD / airflow (later)	M5	SIM-01	To do
Manufacturing outputs				
MFG-01	3D-print prep	M3	MOD-11	To do
MFG-02	CNC CAM	M3	—	To do
MFG-03	Sheet-metal flat DXF	M3	MOD-06	To do
MFG-04	Release package	M2	PDM-05 DRW-08	To do
Robotics & drone kits				
ROB-01	COTS library & insert	M1	MOD-05	To do
ROB-02	Frame & arm templates	M3	MOD-03	To do
ROB-03	Mass, CG & inertia report	M1	MOD-11 ROB-01	To do
ROB-04	Drone balance & thrust-to-weight	M3	ROB-03	To do
ROB-05	Kinematic chains & joint drive	M5	ASM-03	To do
ECAD bridge (KiCad)				
ECD-01	ECAD intake & adopt-vs-build	M1	—	To do
ECD-02	Board STEP into assembly	M3	ECD-01	To do
ECD-03	Board outline round-trip	M3	ECD-01	To do
ECD-04	Placement sync & enclosure fit	M3	ECD-02 ASM-06	To do
AI & automation				
AUT-01	Headless hooks & CLI	M0	—	Done
AUT-02	bruceworx.api	M1	AUT-01	To do
AUT-03	MCP server	M1	AUT-02	To do
AUT-04	Assistant panel	M5	AUT-03	To do
AUT-05	Command line	M5	—	To do
UI & styling				
UIX-01	Theme, pref pack & viewport styling	UI	—	Done
UIX-02	macOS title bar	UI	—	Done
UIX-03	Browser (overlay model tree)	UI	—	In progress
UIX-04	History timeline	UI	—	In progress
UIX-05	Workspaces & tabs (Fusion naming)	UI	—	To do
UIX-06	Toolbar & panels	UI	UIX-05	To do
UIX-07	Task panel polish	UI	—	To do
UIX-08	Matcap shading	UI	—	To do
UIX-09	Fresh UI design & re-render figures	UI	UIX-05	To do

ID	Epic	Mile.	Depends on	Status
Kernel study				
KRN-01	OCCT coupling survey	BG	—	To do
KRN-02	STEP exchange harness	BG	FND-05	To do
KRN-03	Kernel decision	BG	KRN-01 KRN-02	To do

2 Test session — Sunday 19 October 2026 (tonight)

What is being tested. The first build with roadmap work in it: the review fixes (FND-06) and Rectangular/Circular Pattern for bodies and components (MOD-01). Everything below already passes headless (171 PartDesign tests, 19 pattern tests, offscreen dialog and observer checks; steps 14–17 were added with the assembly audit and the part-property schema). Tonight is about what only a person at the screen can judge: does it look right, feel right, and survive real clicking?

Branch	`bw/fnd06-mod01-patterns` (uncommitted; base `696260b`)
Build	`build/release` (FreeCAD 1.1.4-bw.1), already rebuilt
Launch	`pixi run freecad-release` (uses `package/bruceworx/run-freecad.sh`)
Keep open	View → Panels → Report view, so warnings show up

#	Do	Expect	OK
1	Start FreeCAD. Look at the left side of the window before moving the mouse there.	The model tree (Browser) is visible without hovering. *Fix: auto-hide now defaults to off.*	☐
2	Check the workbench tabs. *Needs your pending `param_fallback` fix, which is in the working tree.*	Only the trimmed set (PartDesign, Assembly, Sketcher, Part, TechDraw, Draft, Spreadsheet, Start); the Report view shows "BruceWorx: applied the BruceWorx preference pack".	☐
3	PartDesign: a body with a 10 mm box; Linear Pattern, direction 1 = X, 4 instances, spacing 10; direction 2 = Y, 3 instances, spacing 10.	12 boxes as one 40 × 30 × 10 solid. Before the fix this gave 8.	☐
4	Open an old file that has a two-direction linear pattern (if you have one).	It now shows every row; the pattern repairs itself on recompute.	☐
5	Select a *body* (not a feature) → Rectangular Pattern (PartDesign toolbar, after Polar Pattern, or BruceWorx menu). Change quantity, distance, distance type, Symmetric, then tick Direction 2.	Dialog opens with live preview; every change updates the 3D view at once; parallel directions show a red message instead of breaking.	☐
6	In the same dialog, uncheck two rows in "Instances". Press Cancel. Repeat and press OK.	Unchecked instances disappear. Cancel leaves no trace in the tree; OK leaves one "… pattern" object.	☐
7	Select that pattern object and press Rectangular Pattern again.	The dialog reopens with the saved values and suppressed rows; edits apply to the same object.	☐
8	Move the original body (Transform or Placement).	The pattern follows.	☐
9	Select a body → Circular Pattern. Select a circular edge of a hole on another part, press "Use selection" next to Axis. Try Full, Angle 90 °, Symmetric.	Instances spin about that hole's centre; Angle and Symmetric spread as labelled; the angle box is greyed out for Full.	☐
10	Assembly: insert a bolt part, select it, Circular Pattern, quantity 6, OK. Then add a joint to one of the generated bolts.	Six bolts; the BOM / component count sees 6. *Joint to a generated instance is the one case not tested headless.*	☐

#	Do	Expect	OK
11	Repeat 5 with Copies = Independent, OK.	Separate solids (bodies) or separate links (components) appear; they no longer follow the original.	☐
12	Select a PartDesign *feature* (e.g. a pocket) → Rectangular Pattern.	PartDesign's own Linear Pattern dialog opens (features keep using PartDesign until the upstream rebase).	☐
13	Preferences → BruceWorx: untick "Rectangular and Circular Pattern commands", switch workbench.	The two buttons disappear from the toolbars.	☐
Added after the assembly audit (section 5.1); steps 14 and 16 test the joint changes of section 5.3.			
14	Assembly: two parts joined by a Revolute joint, no limits. *Drag*: click the moving part in the 3D view and pull it round. Then edit the joint: tick Min and Max angle, set 0 and 90 °, press *Pick datums* and click a flat face on each part; OK. Drag again.	Without limits it swings all the way round. With them the panel shows "Now at …" measured between the two faces, and the drag stops at 0 and 90 ° from where the faces line up.	☐
15	After step 10: change the bolt pattern quantity from 6 to 8, then to 4 (keep the jointed bolt within the first 4). Then to 2, so the jointed bolt goes away.	The joint stays on its bolt at 8 and 4. At 2 the joint reports a clear error in the tree, no crash.	☐
16	Put two parts where you want them by hand (Transform). Ground one. Add a Fixed joint between a face of each, first with *Keep parts where they are* unticked, then tick it before OK.	Unticked: the second part jumps so the picked faces line up. Ticked: it goes back and stays exactly where you put it.	☐
17	Part properties: open the Python console (View → Panels) on a part file and run `from bruceworx.partprops import load_schema` `load_schema().apply( App.ActiveDocument.Part)` then give the body a material. In an assembly of it, Create BOM and add columns `.PartNumber`, `.Revision`, `.MaterialName`, `.Mass`.	A "BruceWorx" group in the part's properties (P/N, Revision, Material name and Mass greyed out). The BOM shows the material and the mass; a revision typed as 01 shows as 01, not 1.	☐

Report back. For anything that is not as expected: a screenshot, the Report view text, and the step number. Saving the test document (`File` → `Save`) helps reproduce it here.

Not in tonight's build. Symmetric and suppression for PartDesign *features* (they come with the upstream rebase, `FND-03`); the Fusion-style workspaces and panels (UI track, after M2).

3 Track FND — Foundation and upstream

Every other track depends on a fork that builds reproducibly, ships, and keeps up with FreeCAD. Two foundation decisions changed in the interview. BruceWorx now tracks **upstream main** instead of waiting for the 26.3 tag. Fixes found while reviewing the UI now have an epic of their own.

BW-FND-01 Build & CI on Linux and macOS arm64 — M0 — In progress

Pixi environments and lockfiles for both platforms, ccache in CI, and the self-hosted Apple-silicon runner. `BUILDING-BRUCEWORX.md` and `.github/workflows/bruceworx-ci.yml` exist, and the Linux STEP check and hosted macOS lock are merged (#4). *Acceptance:* a fresh clone builds on both platforms using only the documented commands, and CI is green on both.

Ideas absorbed: FCB-P0-CI-01..06, P0-01, P0-06..08, P0-10

BW-FND-02 Packaging, branding & releases — M0 — In progress

The version string already reads `1.1.4-bw.1`, and the unsigned arm64 `.dmg` script is merged (#3). Remaining: tag a release carrying the dmg plus a source tarball, and decide on signing and notarisation before anyone outside the team installs it. *Acceptance:* the tagged release installs on a clean Mac and opens a Fusion STEP export (`LE-01001`, 200×200×8 envelope).

Ideas absorbed: FCB-P0-02, P0-03, P0-09, P0-11, P0-17, P0-18

BW-FND-03 Track upstream FreeCAD main — M0 — To do

`UPSTREAM-SYNC.md` currently says "stay on 1.1.4, rebase onto 26.3 when it ships". That changes to continuous tracking:

- `upstream-main` stays a fast-forward mirror. A nightly job rebases the `[BW]` series onto it and runs the BruceWorx tests plus the headless acceptance corpus (`FND-05`).
- A conflict or red build opens an issue automatically. The *conflict budget* is the number of upstream files our series touches. It is reported each night and must not grow without a reason recorded in the PR.
- Product releases are still cut from a tested point (`main` after a green nightly), not from arbitrary upstream commits.

Why: the Assembly, TechDraw and FEM work that parity depends on lands upstream continuously. Rebasing once a year makes every conflict expensive at once. *Acceptance:* the nightly rebase runs for two weeks; each conflict is resolved within a working day.

Ideas absorbed: FCB-P1-16..21, P0-12 (replaced)

BW-FND-04 Licences & notices — M0 — To do

FreeCAD is LGPL-2.1-or-later. The roadmap bundles more components: SheetMetal (LGPL-2.1-or-later), OCP, CadQuery and build123d (Apache-2.0), possibly an OpenSCAD binary (GPL-2, shipped as a separate executable) and a fastener library (licence to check [verify]). Each needs a line in `NOTICE` and a third-party licence file in the app bundle. *Acceptance:* every bundled component is listed; each release has a source tarball attached.

Ideas absorbed: FCB-P0-13

BW-FND-05 Acceptance corpus & test harness — M0 — To do

One corpus serves every epic: native and STEP versions of LE parts and assemblies, the 2-DOF arm, the quad frame, and the benchmark models in this document. A single command runs every headless check. Screenshots come from the existing before/after script. The renders in this document already come from headless FreeCAD (`scripts/`), and the harness generalises that. *Acceptance:* `pixi run acceptance` runs all checks in CI in under 15 minutes.

Ideas absorbed: FCB-P1-09, P3-08

BW-FND-06 Fixes found in review — M0 — In progress

1. **Linear Pattern direction 2 drops a row** (§4.1). `Spacings2` is declared `({})`, which stores `[0.0]`. Fix the declaration, migrate saved files, add a test.
2. **Workbench trim not applied.** The pack lists FEM, CAM, BIM, Robot and others as disabled (`bruceworx_pack.py:118`), but the screenshot shows 13 tabs. The uncommitted `param_fallback` change suggests the pack crashed on `int(None)` [verify].
3. **Overlay model tree missing** on the screenshot, although it is on by default; root cause unknown [verify].

Acceptance: a 4×3 pattern gives 12 instances; only intended workbenches are visible; the tree shows on first run.

Ideas absorbed: (new)

BW-FND-07 On by default, each with an off switch — M0 — Done

Merged in #9. It remains a rule for every epic below: a feature ships enabled and can be switched off on Preferences → BruceWorx, and switching it off restores stock behaviour.

Ideas absorbed: FCB-P1-01, P1-10

4 Track MOD — Part modeling

Modeling is the main track and comes first. M1 needs `MOD-01`–`MOD-05`, `MOD-11` and `MOD-12`. Sheet metal and frames serve M3. Surfacing, freeform and mesh conversion are M5.

Capability	FreeCAD 1.1.4 today (checked in this repo)	Gap to Fusion / SolidWorks
Feature patterns	`LinearPattern` (2 directions), `PolarPattern`, `MultiTransform`	Bug in direction 2; no symmetric, no suppress
Body/component patterns	Draft link arrays	Not assembly-aware; no Fusion options
Parameters	Spreadsheet, `App::VarSet`, expressions	No Change Parameters dialog
Top-down	`ShapeBinder`, `SubShapeBinder` (cross-document)	No skeleton/derive workflow
Configurations	none (no configuration table in `src/Mod/Spreadsheet` or `src/App`)	Missing
Direct edit	`Part_Defeaturing` (delete faces)	No press-pull, move or replace face
Threads	`Hole` has `ModelThread`; TechDraw draws cosmetic threads on views	No model-level cosmetic thread
Sheet metal	add-on only	Bundle (decided)
Weldments	none in core	Missing
Surfacing	Surface WB: Filling, Sections, GeomFillSurface, ExtendFace, BlendCurve; Part: Ruled, Offset, Thickness	Trim/untrim, stitch, continuity analysis
Freeform	none	Missing
Mesh → solid	`Part_ShapeFromMesh` (faceted), ReverseEngineering WB	No segmentation/rebuild
Inspect	Measure incl. `MeasureCOM`; `Part_CheckGeometry`	No overhang/draft/thickness maps

4.1 BW-MOD-01 — Rectangular and circular patterns

`BW-MOD-01` Rectangular & circular patterns M1 **In progress**

Two commands under CREATE: Rectangular Pattern and Circular Pattern. Each works on features, bodies and assembly components, with Fusion's options: symmetric direction, suppress instances, and linked vs. independent copies (linked by default). Depends on `FND-06` (the direction-2 fix). *Acceptance:* Figures 4 and 5 reproduced in the GUI; undo/redo; save/reload; the Assembly workbench sees every component instance.

Ideas absorbed: (new, from the UI plan)

Fusion option	FreeCAD 1.1 today	Plan
Pattern type: Features	PartDesign `LinearPattern` (2 directions), `PolarPattern`	Wrap these, fix the direction-2 bug
Pattern type: Bodies	Draft `make_ortho_array` / `make_polar_array` with `use_link`	New command builds a link array, or copies
Pattern type: Components	Assembly understands link arrays (`UtilsAssembly.py:127-131`)	New command builds an `App::Link` array inside the assembly
Distance type: Extent / Spacing	`Mode` = Extent / Spacing	Same
Direction type: Symmetric	—	**New**: index range $-\lfloor\frac{n-1}{2}\rfloor..\lceil\frac{n-1}{2}\rceil$
Circular type: Full / Angle / Symmetric	Angle + Extent/Spacing	Map Full/Angle; **new** Symmetric
Suppress instances	— (features)	**New** for features; links: `setElementVisible`
Linked / independent	n/a	Dialog option (bodies, components)

Theory

Rigid motions. An instance is the seed moved by a rigid motion $g \in SE(3)$,

$$g = \begin{pmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{pmatrix}, \qquad R \in SO(3),\ \mathbf{t} \in \mathbb{R}^3.$$

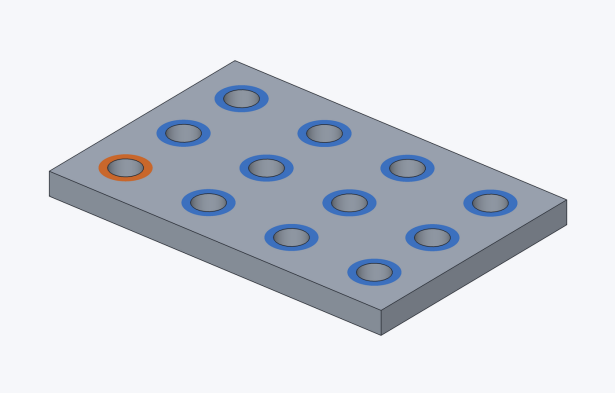
A pattern is a finite family $\{g_k\}_{k\in K}$ with $g_{\mathbf{0}} = I$ (the seed). Features change one solid with Booleans, $S \cup \bigcup_{k\neq 0} g_k(T)$ (additive) or $S \setminus \bigcup_{k\neq 0} g_k(T)$ (subtractive), where S is the support and T the seed's tool shape. Bodies and components only produce *placements* $\{g_k P_0\}$, so no Boolean is evaluated.

Rectangular lattice. With unit directions $\mathbf{u}_1, \mathbf{u}_2$ and per-gap spacings $s_q^{(a)}$,

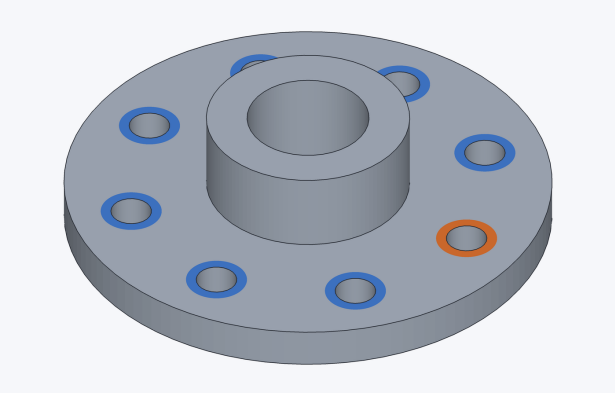
$$\mathbf{t}_{ij} = d_1(i)\,\mathbf{u}_1 + d_2(j)\,\mathbf{u}_2, \qquad d_a(m) = \operatorname{sgn}(m) \sum_{q=1}^{|m|} s_q^{(a)}.$$

In *Extent* mode $s_a = L_a/(n_a - 1)$, and in *Spacing* mode $L_a = (n_a - 1)s_a$. These are `calculateOffsetVector` and `calculateSteps` in `FeatureLinearPattern.cpp`. The index sets are $I_a^{\text{one}} = \{0, \dots, n_a - 1\}$ and $I_a^{\text{sym}} = \{-\lfloor\frac{n_a-1}{2}\rfloor, \dots, \lceil\frac{n_a-1}{2}\rceil\}$, and $K = (I_1 \times I_2) \setminus K_{\text{supp}}$. Translations commute, so the order of the two directions is irrelevant. Instances are distinct iff every $s_q^{(a)} \neq 0$ and $\mathbf{u}_1 \nparallel \mathbf{u}_2$. The bug below breaks the first condition.

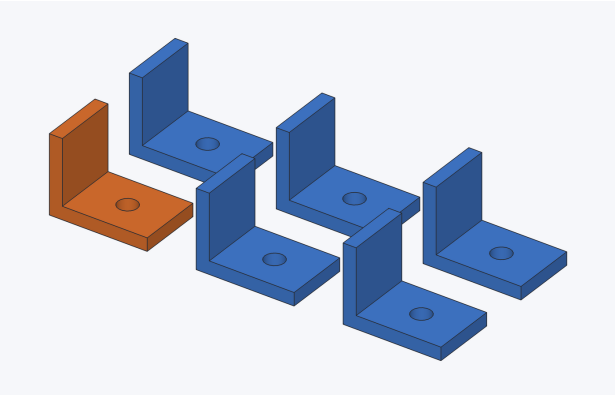
Circular pattern. About an axis through $\mathbf{c}$ with direction $\mathbf{a}$, Rodrigues' formula gives $R(\mathbf{a}, \theta) = I + \sin\theta\,[\mathbf{a}]_\times + (1 - \cos\theta)\,[\mathbf{a}]_\times^2$ and $g_k : \mathbf{p} \mapsto \mathbf{c} + R(\mathbf{a}, \theta_k)(\mathbf{p} - \mathbf{c})$. The schedule is $\theta_k = 2\pi k/n$ for *Full* (dividing by n, since $2\pi \equiv 0$), and $\theta_k = \Theta k/(n-1)$ for *Angle* ($k \in I^{\text{one}}$) and *Symmetric* ($k \in I^{\text{sym}}$). Rotations about different axes do not commute, so nested patterns use a fixed order, $g = g_k^{\text{outer}}\, g_{ij}^{\text{inner}}$.



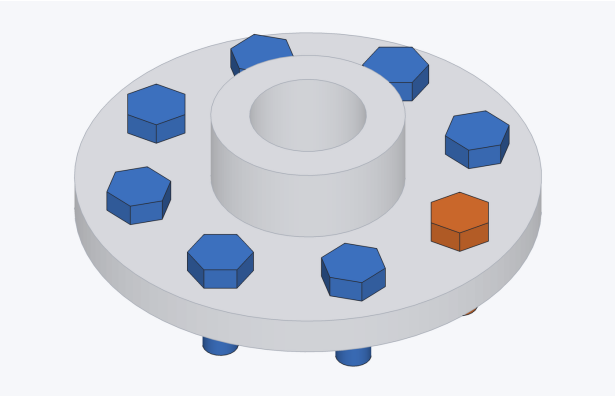
(a) Feature, rectangular: 4×3 holes (`LinearPattern`, two directions).



(b) Feature, circular: 8 bolt holes (`PolarPattern`, Full).



(c) Bodies, rectangular: 3×2 brackets (link array).



(d) Components, circular: 8 bolts around the flange axis.

Figure 4: Pattern examples. Seed in orange, generated instances in blue. Panels (a) and (b) are real PartDesign features recomputed in headless FreeCAD 1.1.4 (panel (a) with the direction-2 workaround). Panel (c) was built with Draft's link array (6 elements confirmed). Panel (d) uses the same transforms as the component command.

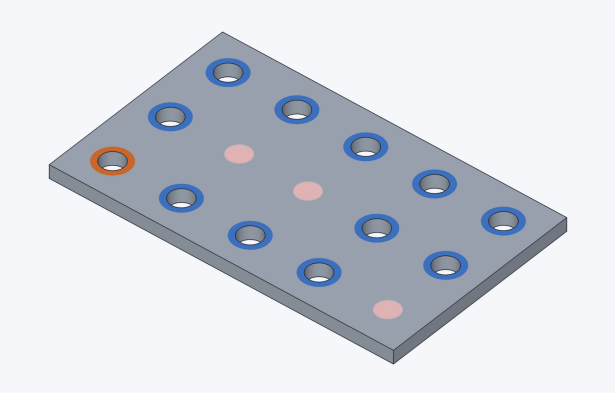
Placements. Pattern transforms are defined in world coordinates, so instance placements are the left product $P_k = g_k P_0$. The right product $P_0 g_k$ would pattern in the seed's local frame.

Linked vs. independent. A link array stores the seed B-rep once plus n placements, $O(|B| + n)$ memory, and follows edits. Independent copies cost $O(n|B|)$. Verified headless: `App::Link` with `ElementCount`$= 8$ produced 8 `App::LinkElements` and a compound of 8 solids. `setElementVisible("3", False)` dropped it to 7, which gives suppression for bodies and components with no new code.

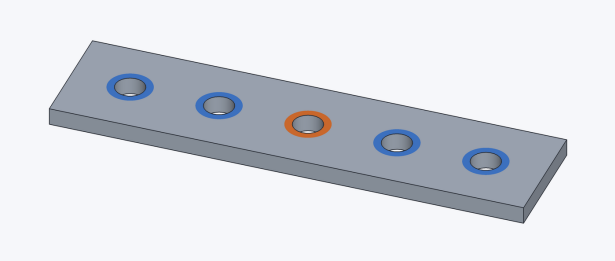
Compute modes. Fusion's *Identical* equals FreeCAD's `TransformMode = Features` (the tool shape is copied). *Adjust* re-evaluates each instance's terminations and needs a re-execute per instance, so it is deferred (§15).

The direction-2 bug.

Probe (headless FreeCAD 1.1.4)	rows asked	rows made	`Spacings2`
Spacing 25 mm, $n_2 = 3$	3	2 ($y = 15, 40$)	`[0.0, -1.0]`
Extent 50 mm, $n_2 = 3$	3	3	`[0.0, -1.0]`
Spacing 25 mm, $n_2 = 5$	5	4	`[0.0, -1.0, -1.0, -1.0]`
Direction 1, Spacing 30 mm, $n_1 = 4$	4	4	(`Spacings` correct)



(a) Suppression: $K = (I_1 \times I_2) \setminus K_{\text{supp}}$, $|K| = 12$.



(b) Symmetric direction, $n = 5$: $I = \{-2, \ldots, 2\}$.

Figure 5: The two new pattern options (target behaviour, not output of existing FreeCAD features).

```
ADD_PROPERTY_TYPE(Spacings,  ({-1.0}), "Direction 1", ...);  // ok
ADD_PROPERTY_TYPE(Spacings2, ({}),     "Direction 2", ...);  // {} -> setValue(double{})
    -> [0.0]
```

The leading 0.0 survives `updateSpacings`, so the first cumulative step is zero and a duplicate row lands on the seed. Fixed on 19 Oct 2026: declared `({-1.0})`, normalised in the constructor (as upstream main now does), and a stored 0 is mapped to −1 and written back, which repairs saved files. Three regression tests (spacing, extent, old-file repair); all 171 PartDesign tests pass. Upstream main still declares `({})`; its constructor masks the bug for new objects but not for saved files, so the repair is worth upstreaming (**FND-06**).

Commands (built 19 Oct 2026, in GUI testing). `BruceWorx_RectangularPattern` and `BruceWorx_CircularPattern` (`bruceworx_pattern.py`, `bruceworx_pattern_gui.py`) dispatch on the selection. *PartDesign features* go to PartDesign's own Linear/Polar Pattern. *Bodies, Part solids and assembly components* get an `App::Link` array whose parameters are dynamic properties in a "BruceWorx Pattern" group: rectangular (two directions, spacing or extent, symmetric) or circular (full, angle, symmetric; axis from X/Y/Z, a straight edge, a circular edge's centre or a datum axis), suppression per instance, and linked or independent copies. A task panel previews live, and Cancel removes everything with or without document undo. Moving the seed re-places the pattern. In an assembly, the array's elements count as components (`UtilsAssembly.isLinkGroup`; verified 6 components for a 6-up bolt pattern). 19 headless tests plus offscreen dialog smoke tests.

Upstream changes the feature path. Upstream `main` (June–October 2026, PRs #30417 and #32843) moved Linear/Polar Pattern logic into `Part::LinearPatternExtension` and friends and added per-instance suppression (`SuppressedPositions`). Symmetric directions and suppression for *features* are therefore taken from upstream after the rebase (**FND-03**), not written again on 1.1.4. Symmetric for features remains a candidate upstream contribution.

4.2 BW-MOD-02 — Parameters and top-down design

`BW-MOD-02` Parameters & top-down design M1 To do

- **Change Parameters** dialog (Fusion's name) listing every user parameter with its expression, value, unit and comment, drawn from `VarSets`, Spreadsheet aliases and named feature dimensions, with search and favourites.
- **Skeleton** workflow: a master sketch or skeleton part drives layout, and parts consume it

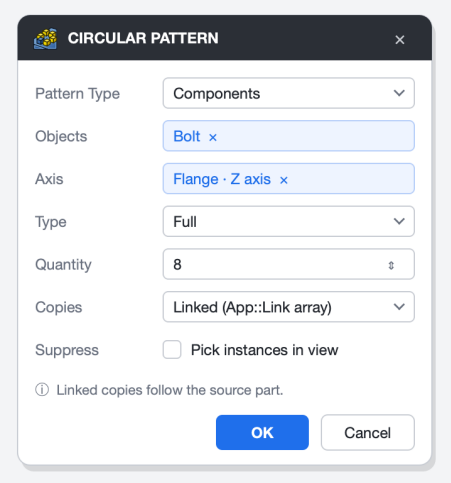


Figure 6: Circular Pattern dialog for components (working mock). *Copies* shows for bodies and components; *Compute* only for features.

through `SubShapeBinder` across files. That fits the one-file-per-part model (`ASM-01`).

- **Derive**: bring selected bodies, sketches or parameters from another file in as a linked, read-only reference, like SolidWorks' Insert Part or Fusion's Derive.

Acceptance: changing a skeleton dimension updates three linked part files in one recompute.

Ideas absorbed: (new)

Theory: the recompute graph. A document is a directed graph $G = (V, E)$ with an edge $u \to v$ when v reads u through a link or an expression. Recompute evaluates V in a topological order of G, in $O(|V| + |E|)$ time, and only for the *touched* set's descendants. Top-down design adds edges *between documents*. The combined graph must stay acyclic. A part that binds geometry from an assembly which contains that same part would close a cycle $p \to a \to p$. The tooling therefore refuses that binding up front, rather than letting recompute fail. It allows a skeleton part referenced by both instead.

4.3 BW-MOD-03 — Configurations

`BW-MOD-03` Configurations M1 To do

One part or assembly file holds several variants (Fusion: Configurations; SolidWorks: configurations and design tables). A **configuration table** object has rows for configurations and columns for parameters, feature suppression states, custom properties (including the part number), and for assemblies, which configuration each component uses. Activating a row writes the values and recomputes. *Acceptance:* Short, Standard and Long brackets come from one file (Figure 7), and the BOM lists each configuration's own P/N.

Ideas absorbed: (new)

Theory. Let $\mathbf{p} \in \mathcal{P}$ be the parameter vector and $\mathbf{s} \in \{0,1\}^F$ the suppression state of F features. A configuration is $c_j = (\mathbf{p}_j, \mathbf{s}_j, \pi_j)$, where π_j holds its properties (P/N, description, material). The model is a function $M(\mathbf{p}, \mathbf{s})$ evaluated by recompute. Configurations change *inputs only*, never topology of the feature graph, so one history serves all rows. Two consequences follow:

1. Each configuration is a distinct item for manufacturing, so it carries its **own P/N**. The BOM resolves a component to (file, c_j), not just to a file.

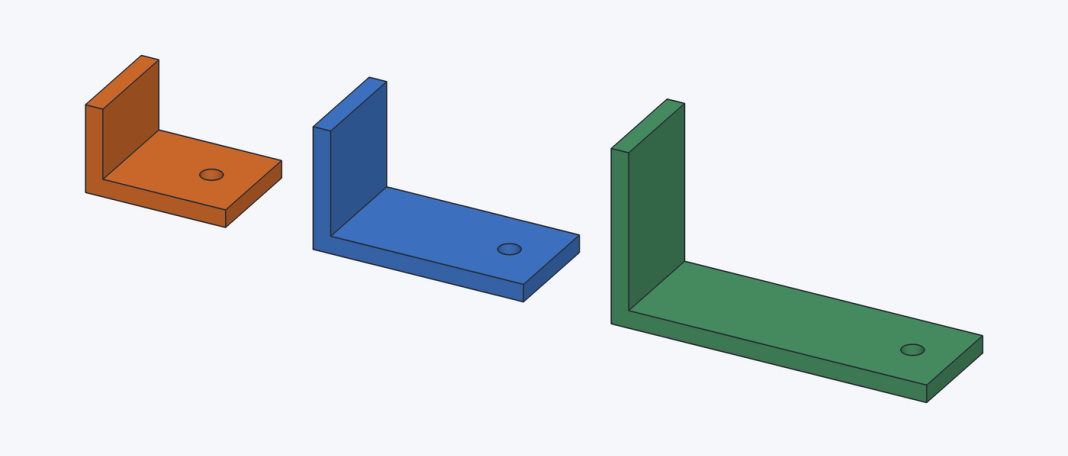


Figure 7: Three configurations of one PartDesign body. A `VarSet` drives `Length`, `Height` and the hole position by expression, and recomputing with three value sets gives volumes of 8 858.6, 13 358.6 and 20 108.6 mm^3. This is the mechanism the configuration table automates. Built by `scripts/render_roadmap.py`.

2. Topological naming must survive a row switch. A fillet on an edge that exists only when a feature is unsuppressed has to be suppressed in the rows where that edge is missing. Per-row validation runs every configuration in CI and reports which features fail in which rows.

Storage: a `BruceWorx::ConfigurationTable` object (FeaturePython) in the part file. It can import and export CSV, so variants can be edited in a spreadsheet, as with a SolidWorks design table.

4.4 BW-MOD-04 — Direct edit on imported geometry

`BW-MOD-04` Direct edit on imported geometry M1 **To do**

Edit "dumb" solids from STEP (Fusion exports, vendor parts) without a feature history: **delete face** (exists: `Part_Defeaturing`), **press-pull / offset face**, **move face** (translate or rotate), **replace face** (with a surface), plus **feature recognition** that turns cylindrical cut faces into editable holes. Each edit is a parametric feature on top of the imported shape, so it appears in the timeline. *Acceptance:* the Fusion STEP of `LE-01001` has a hole moved and a face offset, and stays valid.

Ideas absorbed: FCB-P2-15

Theory: local face offset. Press-pull replaces a face F on surface $S(u, v)$ with its offset $S_d(u, v) = S(u, v) + d\,\mathbf{n}(u, v)$, then re-intersects S_d with the surfaces of the faces adjacent to F to rebuild the edges, and extends or trims those neighbours. That is OCCT's `BRepOffsetAPI_MakeOffsetShape` in local mode, or `BRepFeat` for planar faces. Validity conditions:

- $|d|$ must stay below the minimum radius of curvature of S where d points to the concave side, or the offset self-intersects;
- each neighbour must still intersect S_d. A neighbour that "runs out" (a blend narrower than d) is removed first, which is how Fusion handles fillets that disappear when a face is pushed.

Feature recognition: cylindrical faces sharing an axis, with a closed boundary on a planar face, become a hole candidate $(\mathbf{c}, \mathbf{a}, r, \text{depth})$. Counterbores are coaxial pairs with $r_2 > r_1$. The recognised hole is replaced by a real `Hole` feature, which also feeds drawing callouts (`DRW-04`).

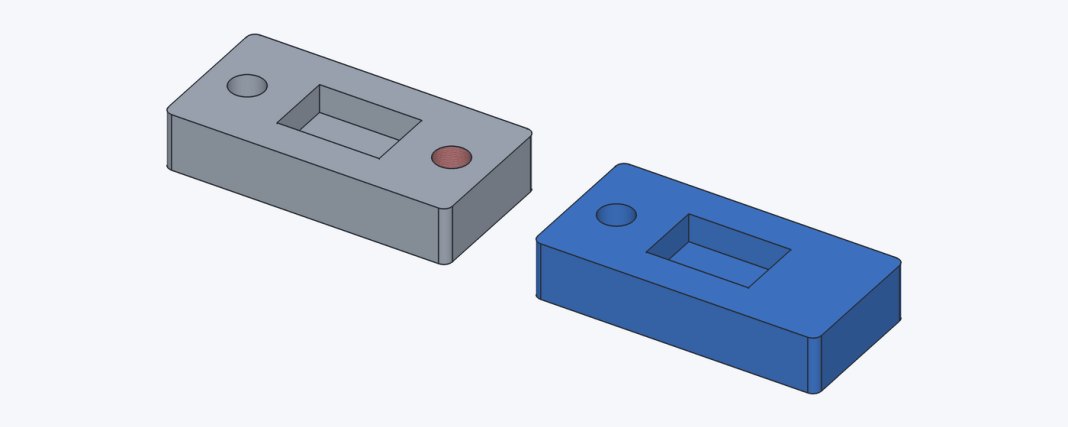


Figure 8: Direct edit available today. A filleted block is exported to STEP and re-imported as a dumb solid (17 faces). The right-hand hole's faces (red) are removed with `Shape.defeaturing` in headless FreeCAD, giving 16 faces and a valid solid (blue). Press-pull, move and replace face build on the same kind of local operation.

4.5 BW-MOD-05 — Threads, holes and hardware

`BW-MOD-05` Threads, holes & hardware library M1 To do

- **Cosmetic thread** object attached to a cylindrical face, storing standard, size, pitch, class, length, and handedness. It is shown in 3D as a texture/helix decoration, exported in STEP AP242 PMI where supported, and read by TechDraw for callouts.
- **Hole presets** (Hole Wizard parity): clearance (close/normal/loose), tapped, counterbore and countersink per fastener, plus **heat-set insert** holes by insert part number, the common case for FDM parts.
- **Fastener library**: evaluate the FreeCAD Fasteners add-on (licence and maintenance [verify]) against a BruceWorx library built on the COTS schema (`ROB-01`).

Acceptance: an M3 tapped hole and an M3 heat-set hole both produce correct callouts on the drawing.

Ideas absorbed: FCB-P2-09/10, (new)

Theory: thread geometry. For ISO metric threads of nominal diameter d and pitch P, the fundamental triangle height is $H = \frac{\sqrt{3}}{2}P$. The basic internal minor diameter is $D_1 = d - 2 \cdot \frac{5}{8}H = d - 1.0825\,P$, and the external minor diameter is $d_3 = d - 1.2269\,P$. Tap drill $\approx d - P$ (M3×0.5 $\rightarrow$ 2.5 mm). A cosmetic thread stores (d, P, class, L) and draws the minor/major diameter circle in views. Modelled threads (`ModelThread`) are kept for 3D printing, where the printed geometry is the thread.

4.6 BW-MOD-06 — Sheet metal

`BW-MOD-06` Sheet metal M3 To do

Bundle the SheetMetal add-on (decided; LGPL-2.1-or-later since v0.4.00) and add per-material, per-thickness **rules** (bend radius, K-factor, relief) as a library, like Fusion's sheet-metal rules. Flats and DXF are covered in `MFG-03`, and the drawing in `DRW-07`. *Acceptance:* the bracket in Figure 9 unfolds to the length given by the bend-allowance formula within 0.01 mm.

Ideas absorbed: (new, from the UI plan)

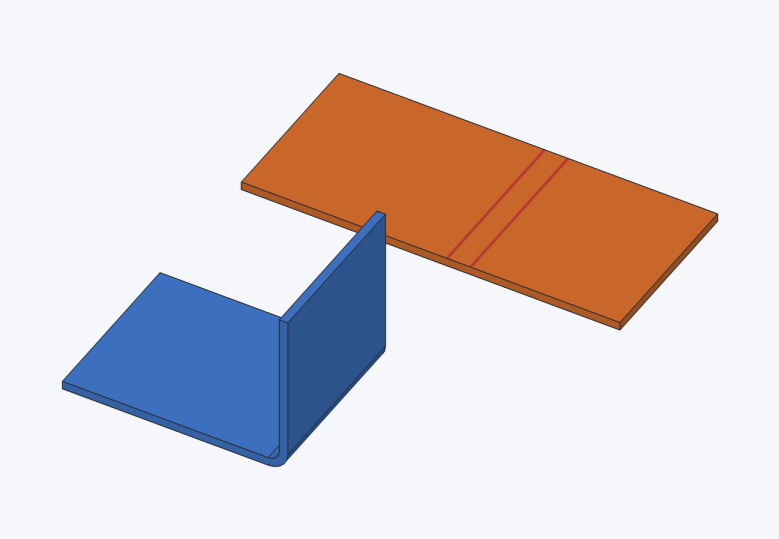


Figure 9: 90° bracket (blue): $T = 1.5$, $R_i = 2$, outside flanges 40 and 30 mm, depth 30 mm, built as one valid solid (volume 3 029.4 mm^3, matching the analytic value). Its flat (orange) is 67.178 mm long, with the bend zone between the red lines ($BA = 4.178$ mm at $K = 0.44$).

Theory: bend allowance. During bending the neutral fibre sits at $R_i + KT$ from the bend axis, with $0 < K < 0.5$ depending on material and process. Its arc length is the bend allowance

$$BA = \theta\,(R_i + K\,T), \qquad L_{\text{flat}} = \sum_i \ell_i + \sum_j BA_j,$$

where ℓ_i are the straight lengths between bend tangent lines. For the bracket, $\ell_1 = 40-(2+1.5) = 36.5$, $\ell_2 = 30 - 3.5 = 26.5$, and $BA = \frac{\pi}{2}(2 + 0.44 \cdot 1.5) = 4.178$, so $L_{\text{flat}} = 67.178$ mm. K comes from the rules library per material, thickness and tooling. That is why the rules table matters more than the unfold algorithm.

4.7 BW-MOD-07 — Frames and weldments

`BW-MOD-07` Frames & weldments M3 **To do**

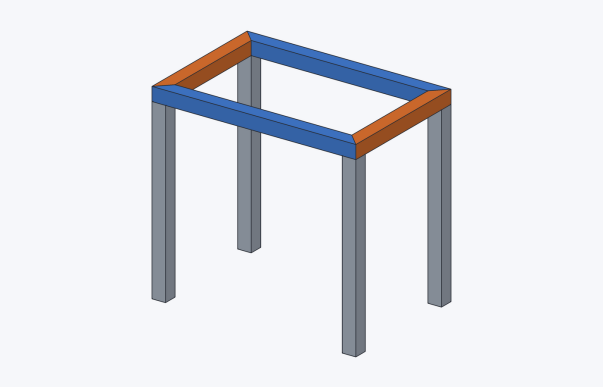
Structural members swept along 3D-sketch edges from a **profile library**: 2020/2040 aluminium extrusion, square and round tube, angle, flat bar. Each joint gets an **end treatment** (mitre, butt one or the other, cope), and **trim/extend** works against other members. The **cut list** groups identical members and gives stock length, angles and quantity. Each member is a body with a cut-list item, not a separate part file. *Acceptance:* the frame in Figure 10 produces the cut list below.

Ideas absorbed: (new)

Theory: mitre plane. Two members meet at corner **c** with unit directions $\hat{\mathbf{d}}_1$ (incoming) and $\hat{\mathbf{d}}_2$ (outgoing). The mitre plane passes through **c** with normal

$$\mathbf{n} = \frac{\hat{\mathbf{d}}_1 + \hat{\mathbf{d}}_2}{\|\hat{\mathbf{d}}_1 + \hat{\mathbf{d}}_2\|},$$

which bisects the turn. Each member is cut by the half-space on its own side. With turn angle $\alpha = \arccos(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)$, the cut is $\alpha/2$ from square, so a 90° corner gives 45°. For profile width w, the outer edge is longer than the centre line by $\frac{w}{2}\tan\frac{\alpha}{2}$ at each mitred end, and the inner edge shorter by the same amount. With $w = 20$ the centre lines are 280 and 180 mm, which gives the 300/260 and 200/160 pairs above.



Member	Qty	Length (outer/inner)	Ends
Top, long	2	300 / 260	45/45
Top, short	2	200 / 160	45/45
Leg	4	260	0/0

20×20 profile. Generated by `render_roadmap.py`; the fused frame is a valid solid.

Figure 10: Mitred top frame on butt-jointed legs, and its cut list.

4.8 BW-MOD-08 — Surfacing parity

`BW-MOD-08` Surfacing parity M5 To do

Fill the gaps next to the Surface workbench: trim and untrim by curve or surface, patch (n-sided, with continuity per edge), extend, stitch into a solid, replace face, and thicken. Add **continuity analysis** (zebra stripes, curvature comb, deviation map), which is how surfacing is checked in Fusion and SolidWorks. *Acceptance:* a surface benchmark set rebuilt to the stated G1/G2 continuity, and thickened into a valid solid.

Ideas absorbed: (new)

Theory: continuity. Two surfaces meeting along a curve are G^0 if positions match, G^1 if tangent planes match ($\mathbf{n}_1 = \pm\mathbf{n}_2$ along the curve), and G^2 if normal curvatures match in every direction. Zebra stripes are reflection lines of a striped environment. A stripe stays continuous across a G^1 joint, and stays continuous *and* smooth across G^2. That makes them a fast visual test of the next-higher order. The tolerances checked are angular ($\leq 0.1°$ for G^1) and relative curvature ($\leq 5\%$ for G^2).

4.9 BW-MOD-09 — Freeform (subdivision / T-spline-like)

`BW-MOD-09` Freeform M5 To do

A **Form body** like Fusion's Form workspace: start from a primitive cage, push and pull faces, edges and vertices, extrude, crease, and mirror. The result converts to a B-spline B-rep solid that the rest of FreeCAD can use. T-splines are proprietary, so the open path is **Catmull–Clark subdivision**, for example via OpenSubdiv (Apache-2.0) [verify], with conversion to bicubic patches. *Acceptance:* a drone canopy modelled as a cage and exported as a valid STEP solid.

Ideas absorbed: (new)

Theory. One Catmull–Clark step creates a face point $\mathbf{f}$ (the face centroid) and an edge point $\mathbf{e} = \frac{1}{4}(\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{f}_1 + \mathbf{f}_2)$, and moves each vertex of valence n to

$$\mathbf{v}' = \frac{\mathbf{F} + 2\mathbf{R} + (n-3)\mathbf{v}}{n},$$

with $\mathbf{F}$ the mean of adjacent face points and $\mathbf{R}$ the mean of adjacent edge midpoints. On regular regions (quads, valence 4) the limit surface *is* a uniform bicubic B-spline. Those patches convert exactly to `Geom_BSplineSurface`. Around extraordinary vertices ($n \neq 4$) the limit surface is not a finite B-spline, so conversion uses G^1 approximating patches (Gregory-type) within a tolerance, and that tolerance is reported. Creases set sharp rules on tagged edges.

4.10 BW-MOD-10 — Mesh to solid

`BW-MOD-10` Mesh to solid M5 To do

A pipeline from STL, 3MF or scan to an editable solid: **repair** (holes, non-manifold edges, flipped normals), **segment** into regions, **fit** planes, cylinders, cones and spheres, then **rebuild** as intersecting analytic faces, or as features where they are recognised. A faceted solid (`Part_ShapeFromMesh`, which exists) remains the fallback. *Acceptance:* a scanned bracket becomes a solid with analytic faces within 0.1 mm of the scan.

Ideas absorbed: (new)

Theory: RANSAC fitting. To fit a primitive needing s points (3 for a plane) to a region where a fraction w of points are inliers, the probability that k random samples all fail is $(1-w^s)^k$. Success with probability p therefore needs $k = \lceil \log(1-p) / \log(1-w^s) \rceil$ samples. For a plane with $w = 0.5$ and $p = 0.99$, that is $k = 35$. Regions are grown from the best model, removed, and the search repeats. Deviation is reported as the RMS and maximum point-to-surface distance.

4.11 BW-MOD-11 — Inspect tools

`BW-MOD-11` Inspect tools M1 To do

Measure (exists), **mass, centre of gravity and inertia** (extends `MeasureCOM`), **section analysis** that stays in the model, **curvature/zebra** (shared with `MOD-08`), **draft analysis**, a **3D-print overhang map** against a build direction, and a **wall-thickness** map. *Acceptance:* mass and CG match a hand-calculated fixture; the overhang map flags the expected faces on a printed part.

Ideas absorbed: FCB-P2R-07, (new)

Theory: overhang. For build direction $\mathbf{b}$ (unit, up) and face normal $\mathbf{n}$, a downward-facing region needs support when its angle from the horizontal is below the critical angle θ_c (typically 45° for FDM), that is when

$$\mathbf{n} \cdot (-\mathbf{b}) > \cos\theta_c .$$

The map colours each tessellation facet by $\mathbf{n} \cdot (-\mathbf{b})$. Draft analysis is the same test against a pull direction with θ_c equal to the draft angle. **Mass properties** are integrals over the solid. For N parts with masses m_i, centres $\mathbf{c}_i$ and inertia tensors I_i about their own centres, the assembly CG is $\bar{\mathbf{c}} = \sum m_i \mathbf{c}_i / \sum m_i$, and the parallel-axis theorem gives $I = \sum_i \left(I_i + m_i(\|\mathbf{r}_i\|^2 E - \mathbf{r}_i \mathbf{r}_i^\top) \right)$ with $\mathbf{r}_i = \mathbf{c}_i - \bar{\mathbf{c}}$ (see `ROB-03`).

4.12 BW-MOD-12 — Sketcher parity audit

`BW-MOD-12` Sketcher parity audit M1 To do

Sketcher is mature, but full parity needs a checked list. Compare it against Fusion and SolidWorks sketch tools: slots, conics, text, offset and project/include, sketch patterns, dimension-driven parameters, constraint inference while drawing, and sketch-level mirror and fillet behaviours. Close the five most-used gaps. *Acceptance:* gap list approved; top five gaps closed.

Ideas absorbed: (new)

5 Track ASM — Assembly

FreeCAD 1.1's Assembly workbench already provides joints (Ondsel solver), grounding, exploded views (`Assembly_CreateView`), a BOM (`Assembly_CreateBom`: Index, Name, File Name, Quantity, plus

custom columns) and kinematic simulation (`Assembly_CreateSimulation`, new in 1.1). This track builds on those. The file model comes first, because PDM, drawings and BOMs all depend on it.

`BW-ASM-01` File model: one part per file — M1 — To do

SolidWorks-style, as decided: every part is its own `.FCStd` with its own P/N, and assemblies reference parts through `App::Link` to external documents with *relative* paths. Tooling: **make external** (split the bodies of a multi-body design file into numbered part files and relink), **pack and go** (copy an assembly with all its references), and a **broken-reference resolver**. *Acceptance:* `LE-02001` opens from a moved folder with no broken links.

Ideas absorbed: (new)

Theory: assemblies as a DAG. Documents form a directed acyclic graph with edges assembly $\to$ component and multiplicities q. A component *occurrence* is a path $a_0 \to a_1 \to \cdots \to p$ from the top assembly. Its world placement is the product of the link placements along the path, $P = P_{a_0a_1} P_{a_1a_2} \cdots P_{a_kp}$. Occurrences, not files, carry joints, colour overrides and BOM item numbers. Making parts external changes nothing in occurrence space, only which document owns the geometry. That is why the conversion can be automatic.

`BW-ASM-02` Assembly gap audit — M1 — Done

Audit current Assembly against a SolidWorks/Fusion checklist and against upstream main. **Approved 19 Oct 2026 (§5.1)**; it re-scopes `ASM-03`–`ASM-10`. Three behaviours still need the screen and are in the test checklist.

Ideas absorbed: FCB-P2A-01/02

5.1 BW-ASM-02 — Assembly gap audit (approved, 19 Oct 2026)

Method. The Assembly workbench in this repository (FreeCAD 1.1.4) was read command by command (`src/Mod/Assembly`), and compared with a SolidWorks/Fusion checklist and with the 97 upstream commits to `src/Mod/Assembly` since the 1.1 release (24 Mar – 10 Oct 2026, GitHub API). The "upstream" column is what the rebase to main (`FND-03`) brings without any BruceWorx work. Items marked [verify] need the GUI and are in the next test session. **Gate:** the mechanical reviewer approves this list; it then re-scopes `ASM-03`–`ASM-10`.

Capability	SW	Fusion	FreeCAD 1.1.4 (this build)	Upstream main since 1.1	Gap → epic
Joints					
Fixed / rigid	●	●	● Fixed	—	—
Revolute, slider, cylindrical, ball	●	●	● all four	joint rotate button	—
Planar	●	●	◑ via Distance on two planes (solver `ASMTPlanarJoint`)	—	UX: name it (ASM-03)
Tangent / cam	●	◑	◑ Distance 0 plane–cylinder = tangent; no cam	—	cam: later
Pin-slot	○	●	○	—	ASM-03
Gear, rack-pinion, screw, belt	●	◑	● Gears, RackPinion, Screw, Belt	coaxial gear joint; rack-pinion fix	—
Parallel, perpendicular, angle, distance	●	◑	●	—	—
Width / symmetric / path / slot mates	●	○	○	—	later (not in Fusion)
Quick joint (type inferred from picks; SW smart mates)	●	●	○ one command per type	"put unconnected part second"	ASM-03

Capability	SW	Fusion	FreeCAD 1.1.4 (this build)	Upstream main since 1.1	Gap → epic
Joint limits	●	●	◑ Length/Angle min–max passed to solver (`ASMTTranslationLimit`, `ASMTRotationLimit`); honoured while dragging (verified headless); not by a plain solve. Zero from datums: built, section 5.3	—	ASM-03
As-built joints (no move)	◑	●	● *Keep parts where they are* (built 19 Oct)	—	ASM-03
Rigid groups	●	●	○	**RigidGroup (#29605)**	rebase
Edit joint from context menu	●	●	◑ double-click	**"Edit joint" entry**	rebase
Structure					
Insert existing part	●	●	● `Assembly_InsertLink`	label fix	—
New part in context	●	●	● `Assembly_InsertNewPart` creates *and saves a separate document* — matches one-file-per-part	joint creation fix	ASM-01 builds on it
Sub-assemblies, rigid vs flexible	●	●	● `AssemblyLink.Rigid`	parent refresh; rigid-change fix	ASM-05 shrinks to verification
Grounding	●	●	●	**grounded parts unmovable**	rebase
Component patterns	●	●	● BruceWorx link arrays (MOD-01, today)	**generated arrays + suppressed elements**	align on rebase
Mirror components	●	●	○	—	new, small (ASM-03)
Position snapshots / configurations	●	●	○	**snapshots (#25384)**	rebase
Motion and checks					
Free drag with solver	●	●	● `preDrag/doDragStep`	island drag (#30998); transformed-assembly fixes	rebase
Kinematic simulation / drive joints	●	●	● `Assembly_CreateSimulation` (Angular/Linear motions, formulas)	formula fixes; **video export (PyAV)**	ROB-05
Dynamic motion (gravity, forces, contact)	●	◑	○	—	ASM-10
Collision detection during drag (Fusion contact sets)	●	●	○	—	ASM-06 (+ later)
Interference / clearance check	●	●	○ nothing in `src/Mod/Assembly`	—	ASM-06
Degrees of freedom readout	●	◑	◑	DoF fix with bundled fixed joints	—
Documentation output					
Exploded views	●	●	● `Assembly_CreateView`	**move controls (#30955)**	rebase + ASM-04 presets
BOM	●	●	◑ Index, Name, File Name, Quantity automatic; **custom columns read same-named properties of each part** (`BomObject.cpp:283`)	—	ASM-04 = add P/N, Rev, Material, Mass *properties* to parts (PDM-02)
Mass properties of assembly	●	●	◑ `MeasureCOM`	—	MOD-11, ROB-03
Scale					
Large-assembly / lightweight mode	●	◑	○	—	ASM-08
In-context editing	●	●	◑ cross-document `SubShapeBinder`	—	ASM-07
Smart fasteners	●	◑	○	—	ASM-09

What the audit changes.

1. **Rebase first, then build.** Six rows close by rebasing (rigid groups, snapshots, exploded-view controls, edit-joint menu, grounded locking, island drag), and upstream now handles generated arrays and suppressed link elements. `FND-03` moves ahead of `ASM-03`.

2. **ASM-04 gets smaller.** The BOM already reads any property by column name, so P/N, Rev and Material columns need only the PDM properties on each part (PDM-02). Only *Mass* needs new code: a computed property kept current on recompute.
3. **ASM-05 becomes verification.** Rigid/flexible sub-assemblies exist (AssemblyLink.Rigid); the epic checks them on LE-02001 and fixes what breaks.
4. **ASM-03 keeps:** quick joint with type inference, pin-slot, mirror components, explicit Planar naming, limits verified while dragging, as-built behaviour.
5. **Still entirely missing:** interference/clearance (ASM-06), dynamic motion (ASM-10), lightweight mode (ASM-08), smart fasteners (ASM-09) — unchanged.

For the next GUI session (steps 14–16 of section 2).

(a) Limits while dragging. *Dragging* is moving a part with the mouse in the 3D view after it is jointed; the solver keeps the joints as you pull. Fusion and SolidWorks stop the part at a joint's limit. FreeCAD passes the limits to the solver; whether a drag stops at them is what the step records.

(b) Joints on pattern instances. Required (answer, 19 Oct): a joint on a generated bolt must survive a change of pattern count while that bolt still exists, and fail with a clear message when it goes.

(c) Fixed joint placement. A joint is made by picking a face or edge on each part; the solver then moves one part so the two picks line up. "Stays put" asks whether a Fixed joint between two parts you have already placed keeps them exactly where they are (Fusion calls this an *as-built* joint), or snaps the second part so the picks coincide. Both are useful; the step records which FreeCAD does today, and ASM-03 adds the other.

Answered later on 19 Oct. (a) Drag stays free; limits only when set in the joint, measured from a datum. (c) Joints keep moving one part to line up the picks; a checkbox keeps parts where they are. Both are built (section 5.3); steps 14 and 16 now test them on screen.

Decision (19 Oct). Audit approved. Part properties must stay open to growth: they now come from a layered schema instead of code (section 5.2).

5.2 Part properties that can grow (built 19 Oct 2026)

Requirement. "Make it so we can expand properties of parts later." The properties every part carries (P/N, revision, description, material, mass, finish, vendor…) are therefore data, not code: a list in JSON that is read in layers. Adding a property means adding one line to a layer file; FreeCAD, the BOM and the vault pick it up without a new build.

Layer (read in this order)	Holds
1. Built in: addons/BruceWorx/config/part_properties.json	The nine BruceWorx properties below. Ships with the build; not edited by hand.
2. Company or vault: e.g. <vault>/part_properties.json	What your company adds: supplier, lead time, RoHS, cost centre…
3. User: ~/.config/bruceworx/part_properties.json	Personal additions while trying something out.

Property	Type	BOM column	Comes from	Needed at
`PartNumber`	text	`.PartNumber`	vault (read-only)	—
`Revision`	text	`.Revision`	vault (read-only)	—
`Description`	text	`.Description`	you	submit
`ItemType`	choice	—	you (part, assy, COTS, dwg)	—
`MaterialName`	text	`.MaterialName`	computed: bodies' material	submit
`Finish`	text	`.Finish`	you	—
`Mass`	mass	`.Mass`	volume × density	—
`Vendor`, `VendorPartNumber`	text	—	you	—

Rules that keep it safe to grow.

- A later layer may *add* properties and may change an existing one's description, default, BOM flag or the stage that needs it. It may *not* change a property's type; that would make old files unreadable, so the layer is refused with a message naming both files.
- Renames are declared (`"renamed": {"Old": "New"}`); applying the schema moves the value.
- Applying never deletes a property it does not know, so a file made with a bigger schema keeps its extra values when a smaller one opens it.
- A clash with a property FreeCAD already has is reported and left alone. The first real run found one: `App::Part` already has a `Material` link, so ours is `MaterialName`.
- "Needed at" feeds the lifecycle: `submit` lists what is still empty before a part can go to review.

BOM. The Assembly BOM prints any property whose column is named `.Property`, so the schema just lists those names. Running it for real exposed an upstream bug: the BOM spreadsheet turned the text "01" into the number 1, so revision 01 printed as "1" (and a P/N with leading zeros would lose them). `BomObject.cpp` now stores text values as text, the same way it already stores the Index column.

Verified headless. 12 unit tests (layers, renames, clashes, required stages, enum choices), and a FreeCAD run on an aluminium plate: mass 135 g from its 50 000 mm^3 and Aluminum-6061-T6 density; material name taken from the body; the BOM row reads `LE-01-01001-001 | 01 | Base plate | Aluminum-6061-T6 | Clear anodise | 135 g`. The Assembly test suite still passes.

Not yet. A FreeCAD command and panel to apply and edit the properties (today: one line in the Python console, step 17 of the checklist), and recomputing Mass whenever the body changes (today: whenever the schema is applied). Both belong to **`ASM-04`** and **`PDM-04`**.

5.3 Joints: drag, limits from datums, keep parts in place (built 19 Oct 2026)

Your answers (19 Oct).

- Drag must work, with no angle limit unless one is set in the joint.
- A limit needs a datum to be measured from.
- A new joint moves the parts so the picks line up; a checkbox makes that optional.

What FreeCAD 1.1 already did (measured headless). Limits are off on a new joint. Dragging is free without them. With limits on, the solver *does* stop a dragged part at them: a 0–90 ° revolute, dragged to 130 °, stops at 90.000 °. A plain solve does not: a part turned to 120 ° by hand stays there. What was missing is the reference. FreeCAD measures the angle between the x axes of the two joint connectors, and the x axis comes from whichever face or edge was picked. So "0 °" is wherever the geometry happens to put it.

Limits from datums. In the joint panel, *Limits → Measured from: Pick datums*, then pick a planar face, a straight edge, or a datum plane or line on each part. The limits are then measured from the position where the two datums line up. A door limited to 0–110 °, with the door's face

and the wall's face as datums, reads 0 when it is closed flush. The panel shows the current reading ("Now at 37.0 °"). With no datums, limits are measured from the joint's own axes, as before. A datum may be given on one side only; the other side then uses its connector's x axis.

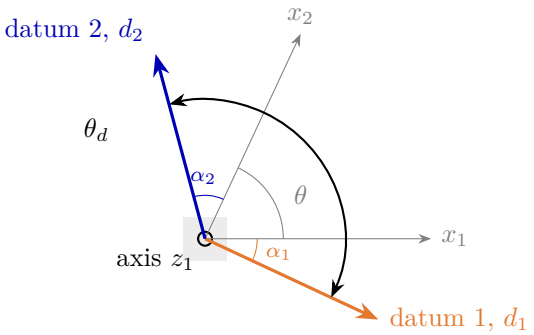


Figure 11: Looking down the joint axis. Each datum sits at a fixed angle α_i in its own connector's frame, because datum and connector are on the same part. The limit is applied to θ_d, the angle from datum 1 to datum 2.

The arithmetic. Turning part 2 by the joint angle θ turns datum 2 with it, so the datum angle is

$$\theta_d = \theta + \alpha_2 - \alpha_1 \qquad \Longrightarrow \qquad \theta = \theta_d + \theta_0, \quad \theta_0 = \alpha_1 - \alpha_2.$$

Here α_i is the angle of datum i's direction, projected onto connector i's xy plane. The direction is a face's normal, an edge's direction, or a datum's own z. Each α_i depends only on where the datum sits on its own part, so θ_0 does not change as the parts move. It is stored on the joint as `LimitAngleZero` (read-only). A limit $\theta_d \in [\theta_{\min}, \theta_{\max}]$ becomes $\theta \in [\theta_{\min} + \theta_0, \theta_{\max} + \theta_0]$ for the solver. Lengths work the same way, with the datums' heights h_i along the axis: $L_0 = h_1 - h_2$, stored as `LimitLengthZero`.

Drag backstop. The solver stops at limits today, but nothing checked that it does. After each drag step, BruceWorx now reads every limited joint itself. If a step would take a joint further past a limit, it is cut back: the move is halved six times from the last good position, so the part stops within 1/64 of one mouse step of the limit. A plain solve that leaves a joint outside its limits now prints a warning in the Report view ("Hinge is at 120 deg, outside its limits (0 deg to 90 deg)"). Python can now drive a drag (`preDrag`, `doDragStep`, `postDrag`), which is how all of this is tested without a screen.

Keep parts where they are. New checkbox in the joint panel, off by default, so a joint still moves one part to line up the picks. Ticked, the joint is fitted to where the parts are: connector 2's offset takes up the difference, $O_2' = O_2\, P_2^{-1} P_1$, where P_i are the two connectors' world placements. Nothing moves, and the joint still works: a revolute made this way turns about connector 1's axis. For a revolute, the angle then reads 0 where the parts were placed, so limits without datums count from the position the parts were built in (Fusion's *as-built* joint). The setting is stored on the joint (`KeepPosition`); ticking or unticking it while the panel is open redoes the joint from the parts' original places. It is shown for Fixed, Revolute, Cylindrical, Slider and Ball joints.

Verified headless. 7 new Assembly tests (`AssemblyTests/TestBWJoints.py`), all passing with the 8 existing ones:

- a free drag to 170 ° and −120 ° without limits;
- a 0–90 ° limit stops a drag at 90 and at 0;
- the reading equals the face-to-face angle worked out from the geometry, at five angles;
- a 0–30 ° limit from two face datums stops a drag at 30 and at 0 measured face to face;

- no datums gives zero offsets;
- a joint moves the part by default, and with *Keep parts where they are* it does not move it, even after a solve, and still turns about the joint.

The panel itself (checkbox, *Pick datums*, reading) needs the screen: steps 14 and 16 of section 2.

Still in `ASM-03`. Quick joint with type inference, pin-slot, limit sliders in the 3D view, datums on the assembly origin or a third part (today both datums are on the joint's own two parts), and continuous angles past ±180 ° (today a limit range lies within one turn).

`BW-ASM-03` Joint UX & limits — M1 — In progress

A pick-two-features **quick joint** that infers the joint type (cylinder + cylinder → revolute, plane + plane → planar, and so on), keyboard shortcuts, and **limit sliders** that the solver honours while dragging. **Built 19 Oct (§5.3):** limits measured from datums, a drag backstop at the limits, and *Keep parts where they are*. *Acceptance:* the 2-DOF arm is assembled with quick joints, and limits stop the drag.

Ideas absorbed: FCB-P2A-03/04

`BW-ASM-04` Exploded views & BOM — M1 — In progress

Exploded-view presets, and a BOM with **P/N, Rev, Description, Material, Mass, Qty**, in *indented*, *top-level* and *flattened (parts only)* forms, exported to CSV. Configurations are resolved per component (`MOD-03`). The same BOM feeds the drawing table (`DRW-02`) and Ops (`PDM-06`). **Started 19 Oct 2026:** the columns come from the part-property schema (§5.2), with Mass and Material computed. *Acceptance:* the `LE-02001` BOM matches a hand count; the flattened BOM sums quantities.

Ideas absorbed: FCB-P2A-05/06

Theory: BOM explosion. With edge multiplicities q_{uv} in the assembly DAG, the total quantity of part p under top assembly t is the sum over all paths of the product of multiplicities,

$$Q_t(p) = \sum_{\pi:\, t \rightsquigarrow p} \prod_{(u,v)\in\pi} q_{uv},$$

computed in $O(|V| + |E|)$ by dynamic programming in topological order, $Q(v) = \sum_{u \to v} Q(u)\, q_{uv}$ with $Q(t) = 1$. The *indented* BOM lists paths, and the *flattened* BOM lists $Q_t(p)$ for leaf parts. *Where-used* (`PDM-06`) is the same traversal on the reversed graph.

`BW-ASM-05` Sub-assemblies — M1 — To do

Rigid vs. flexible sub-assemblies. A flexible one exposes its internal joints to the parent's solver; a rigid one moves as one body. A fix-up macro converts between them. *Acceptance:* a flexible sub-assembly articulates inside `LE-02001`; a rigid one does not.

Ideas absorbed: FCB-P2A-07

`BW-ASM-06` Interference & clearance — M1 — To do

List every **clash** (positive common volume), **contact** (touching faces, zero volume) and **clearance violation** (minimum distance below a threshold) between occurrences. Results are highlighted in the view, with an ignore list for intended press fits and threads. The ECAD enclosure fit (`ECD-04`) reuses it. *Acceptance:* the bolt example in Figure 12 reports one clash, 4.0 mm deep and 78.5 mm^3.

Ideas absorbed: FCB-P2A-08

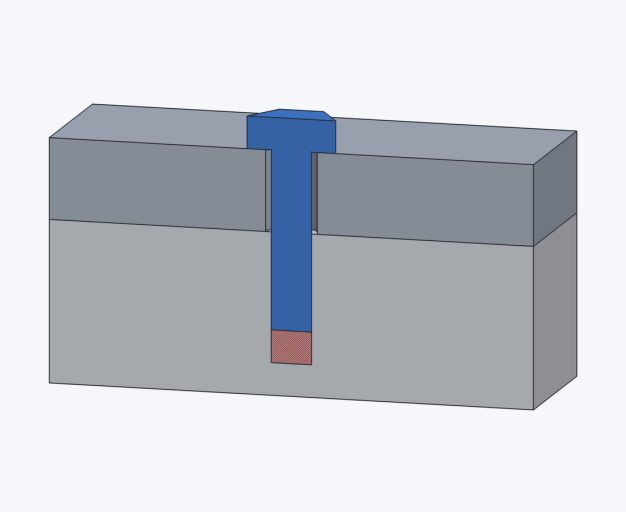


Figure 12: Half section of an M5 bolt that is 4 mm too long for its 12 mm blind hole. The clash volume (red) is the Boolean common of bolt and block: $78.54\,\mathrm{mm}^3 = \pi \cdot 2.5^2 \cdot 4$, computed in headless FreeCAD. The section itself is the kind of view `MOD-11` keeps in the model.

Theory: broad and narrow phase. Checking all $\binom{n}{2}$ pairs exactly is too slow for a robot with hundreds of parts. A *broad phase* sorts the axis-aligned bounding boxes on each axis and sweeps, reporting only pairs whose intervals overlap on all three axes. That takes $O(n \log n + k)$ for k candidate pairs. The *narrow phase* runs the exact Boolean common (clash volume) or `distToShape` (clearance) only on candidates. Contacts are separated from clashes by a volume tolerance ε_V and a distance tolerance ε_d. Pairs on the ignore list (threads, press fits) are filtered by occurrence path, so they stay ignored after the assembly is rebuilt.

`BW-ASM-07` In-context editing — M5 — To do

Edit a part while it is shown inside the assembly, referencing other parts' geometry (Fusion's in-place edit, SolidWorks' in-context features). References become cross-document `SubShapeBinders` and update when the referenced part changes. They can be **locked** (frozen) or **broken**. A dashboard lists all external references. *Acceptance:* bracket holes follow a moved motor mount across files.

Ideas absorbed: (new)

`BW-ASM-08` Large-assembly mode — M5 — To do

Open a full robot quickly. That means **lightweight** loading (display data without the feature tree), **selective open** (choose the sub-assemblies to load), cached tessellation, and level of detail for small parts. The spike measures FreeCAD's existing partial-document loading for links first [verify]. *Acceptance:* an assembly of more than 500 parts opens within a target set by the spike.

Ideas absorbed: (new)

`BW-ASM-09` Smart fastener mating — M5 — To do

Select a hole pattern and a fastener type. The command recognises the holes (`MOD-04` recognition, `MOD-05` data), picks size and length, inserts linked fasteners (and washers and nuts), and joints them. This matches SolidWorks Toolbox smart fasteners and its 2026 fastener recognition. *Acceptance:* bolting a motor mount places and joints all its screws in one command.

Ideas absorbed: FCB-P2R-04

Theory: length selection. For a through-bolt with clamp (grip) thickness g into a tapped or insert hole of usable depth h, the required engagement is $e \geq k\,d$, with $k \approx 1.5$ in aluminium and plastics, or the insert length for heat-set inserts. The chosen standard length is the smallest catalogue length L with

$$g + e \;\leq\; L \;\leq\; g + h - c,$$

where c is the bottom clearance. If the set is empty, the command reports the conflict instead of choosing. Figure 12 shows the failure it prevents.

`BW-ASM-10` Dynamic motion study — M5 — To do

Beyond kinematic animation: **gravity, applied forces and torques, motor profiles, springs and dampers, and contact**, integrated over time, with plots (position, velocity, reaction forces, motor torque) and video export. The spike checks how much of the Ondsel (MbD) solver's dynamics can be used [verify]. *Acceptance:* the arm swinging under gravity matches the analytic pendulum period within 2 %.

Ideas absorbed: FCB-P2A-10

Theory: constrained multibody dynamics. With generalised coordinates $\mathbf{q}$, mass matrix $M(\mathbf{q})$, applied forces $\mathbf{Q}$ and joint constraints $\Phi(\mathbf{q}, t) = \mathbf{0}$,

$$M\ddot{\mathbf{q}} + \Phi_{\mathbf{q}}^{\top}\boldsymbol{\lambda} = \mathbf{Q}, \qquad \Phi(\mathbf{q}, t) = \mathbf{0},$$

an index-3 differential-algebraic system. The Lagrange multipliers $\boldsymbol{\lambda}$ are the joint reaction forces, which is what users want plotted to size motors. It is solved by index reduction to $\Phi_{\mathbf{q}}\ddot{\mathbf{q}} = -(\Phi_{\mathbf{q}}\dot{\mathbf{q}})_{\mathbf{q}}\dot{\mathbf{q}} - 2\Phi_{\mathbf{q}t}\dot{\mathbf{q}} - \Phi_{tt}$ with Baumgarte stabilisation, or with a generalised-α integrator on the index-3 form. Validation uses a pendulum, $T = 2\pi\sqrt{I_O/(mg\ell)}$.

6 Track DRW — Drawings (ASME Y14.5, third-angle)

TechDraw already covers projection groups, section, detail and broken views, dimensions (including chain and coordinate extensions), balloons, weld and surface-finish symbols, template autofill, and PDF/DXF export. The gaps are semantic. GD&T frames are SVG symbols, not data. The BOM table is a spreadsheet view. Thread callouts are drawn on the view, not read from the model. Nothing knows about revisions. This track runs **in parallel from the start** and closes at M2.

`BW-DRW-01` ASME templates & title block — M2 — To do

ANSI A–D sheet templates with the third-angle symbol, a general tolerance block, and fields for P/N, revision, title, material, finish, mass, scale, sheet and approvals. Fields fill from PDM item metadata (`PDM-02`) through TechDraw's existing autofill, which is extended for the new keys. The `LE-01001` sample drawing (overall size, thickness, print Z, PETG/ABS, heat-set inserts) is the first user. *Acceptance:* the `LE-01001` drawing fills every field automatically.

Ideas absorbed: FCB-P2-06/07/08

`BW-DRW-02` Associative BOM table & balloons — M2 — To do

A native TechDraw **parts list** object (ASME Y14.34 columns: item, P/N, rev, description, material, qty) bound to the assembly BOM (`ASM-04`), replacing the spreadsheet view. Balloons bind to *occurrences*, so their numbers follow the item numbers when rows are added, sorted or removed. *Acceptance:* adding a part updates the table and keeps every balloon number correct.

Ideas absorbed: FCB-P2A-06

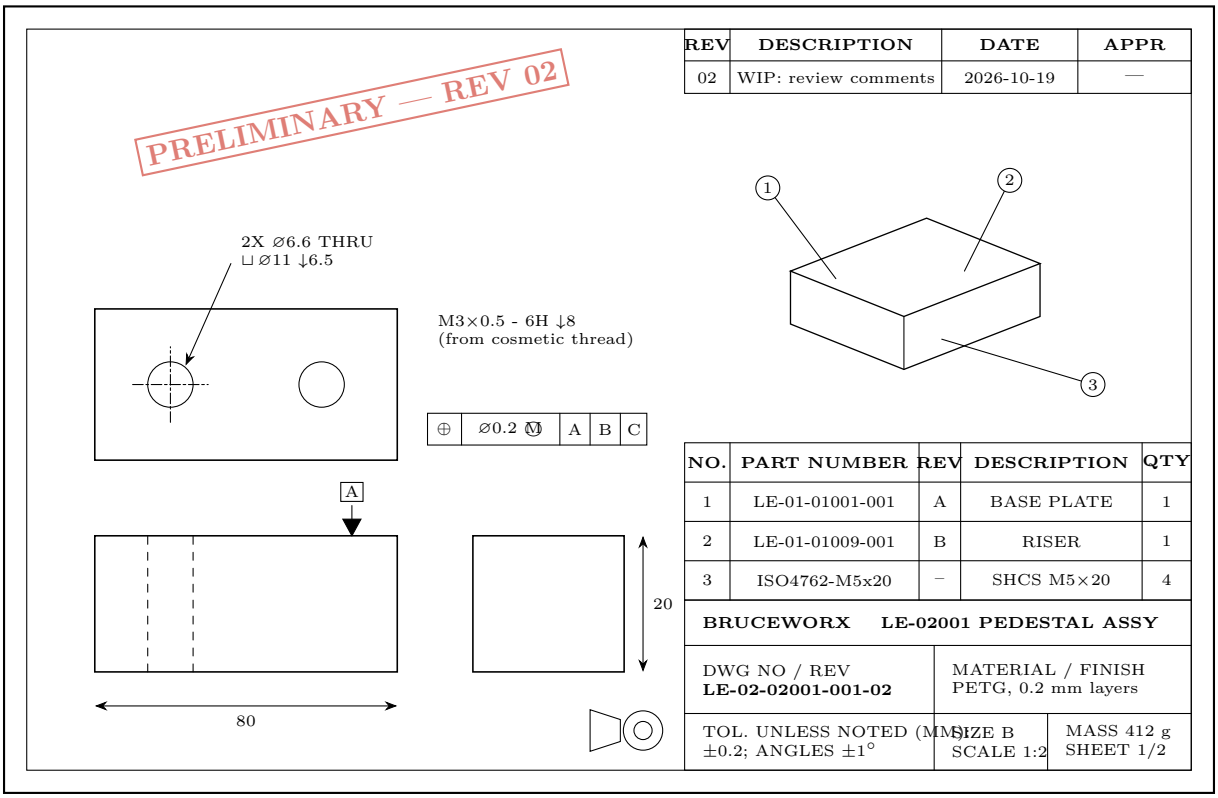


Figure 13: Target ASME B-size sheet (schematic). Third-angle views and symbol; hole and thread callouts generated from model data; a semantic feature control frame against datums A–C; balloons tied to the parts list; revision block and PRELIMINARY stamp driven by the PDM state (numeric rev 02 = not yet released); title block filled from item metadata, printed as `P/N-REV`.

`BW-DRW-03` GD&T & datums (ASME Y14.5) M2 To do

Feature control frames, datum feature symbols, datum targets and basic dimensions as **objects** that reference model geometry. They are validated (every referenced datum exists, modifiers are legal for the characteristic), and they survive model edits through TNP-safe references. *Acceptance:* a frame referencing a missing datum is flagged; frames survive a model change.

Ideas absorbed: (new)

Theory: a feature control frame is a sentence. Y14.5 frames follow a small grammar:

```
fcf        := characteristic zone ( '|' datumRef ){0,3}
zone       := [ 'DIA' | 'SDIA' ] value [ material ] [ 'P' value ] [ 'F' | 'ST' | 'T' ]
material   := '(M)' | '(L)'          ; MMC / LMC (RFS is the default)
datumRef   := letter [ material ] ( '-' letter [ material ] )*
```

Validation rules: form tolerances (flatness, straightness, circularity, cylindricity) take *no* datums; orientation tolerances need at least one; position needs datums unless the frame is a pattern-locating tolerance that is explicitly datum-free; and the material modifier Ⓜ applies only to features of size. Storing the frame as a parsed structure, not as text, is what makes these checks possible. It also allows STEP AP242 semantic PMI export later.

`BW-DRW-04` Hole & thread callouts from the model M2 To do

Generate callout text from `Hole` features and cosmetic threads (`MOD-05`): quantity, diameter, THRU or depth, counterbore ⌴, countersink ∨, thread size, pitch, class and depth. Callouts update when the hole changes. Hole tables (ordinate positions) come from the same data.

Acceptance: callouts update when the hole spec changes.

Ideas absorbed: (new)

BW-DRW-05 Revision table & release stamp — M2 — To do

The revision block is filled from the item's change history (`PDM-05`): ECO number, description, date and approver per letter revision. A **PRELIMINARY** stamp shows while the revision is numeric (01, 02, . . .). The title block prints `P/N-REV`. *Acceptance:* releasing rev A removes the stamp and adds the revision row.

Ideas absorbed: (new)

BW-DRW-06 Auto-generate drawing — M2 — To do

One command makes a first-pass drawing: sheet size and scale, the standard views, an isometric, hole callouts, and the key dimensions. The user then edits it. SolidWorks 2026 ships an equivalent, AutoGenerate Drawing. *Acceptance:* on the corpus parts, a reviewer keeps at least 70 % of the generated dimensions.

Ideas absorbed: (new)

Theory: choosing views and dimensions.

1. **Front view.** Among the six principal directions $\pm\mathbf{e}_x, \pm\mathbf{e}_y, \pm\mathbf{e}_z$ of the part's oriented bounding box, choose the one maximising $\alpha\, A_{\text{proj}} + \beta\, N_{\text{visible edges}} - \gamma\, N_{\text{hidden edges}}$, with ties broken by the longest extent horizontal. The other views follow from third-angle projection: top above, right to the right.
2. **Needed views.** Add a view only if it shows features (holes, steps) whose axis is perpendicular to it and which no earlier view shows true size. A part with all holes on one axis gets two views, not three.
3. **Scale and sheet.** The smallest standard sheet (A–D) and the largest standard scale such that the view group fits within 80 % of the drawing area.
4. **Dimensions.** Overall extents first, then hole positions as ordinate or baseline dimensions from the datum features (the largest planar faces, in order), then sizes through callouts. Nothing is dimensioned twice: each model parameter maps to at most one dimension.

BW-DRW-07 Sheet-metal flat & cut-list drawings — M3 — To do

A flat-pattern view with bend lines, direction and angle, a bend table (bend, direction, angle, radius, K-factor), and the weldment cut-list table from `MOD-07`. *Acceptance:* the bracket drawing in Figure 9 shows the flat and the table.

Ideas absorbed: (new)

BW-DRW-08 Publish PDF/DXF — M2 — To do

Vector PDF and DXF export of every drawing in a BOM, with file names `P/N-REV.pdf`. It feeds the release package (`MFG-04`). *Acceptance:* one command publishes every drawing of `LE-02001`.

Ideas absorbed: (new)

7 Track PDM — Part and drawing management

FreeCAD has no PDM: no vault, check-out, revision, lifecycle, where-used or release. Fusion has versions in its cloud plus the Manage extension (part numbering, lifecycle, change orders, BOM release). SolidWorks has PDM Standard and Professional (vault, check-in/out, workflows, revisions,

BOMs). Searches for an open-source PDM with a working FreeCAD check-in workflow found only early or narrow projects (FreePDM, a concept; a small Taack-based PLM; LibrePLM, with a Windows-only connector). This track therefore **starts with an evaluation**, as decided, and specifies the data model and lifecycle independently of where they end up stored.

`BW-PDM-01` PDM evaluation & decision M2 **Done**

Compare four architectures against weighted criteria and record the decision. **Decided 19 Oct 2026 (§7.1):** a vault in any folder, iCloud Drive first, opened through a `.bwvault` ref file; one writer at a time for now. The folder layout is designed so a sync service has nothing to conflict over.

Ideas absorbed: (new)

7.1 BW-PDM-01 — Decision: a vault in a folder (19 Oct 2026)

Decision (your answer, 19 Oct). The vault lives in any folder you choose and builds itself there; a ref file lets a reader view and drive it; the backend is that folder, most likely in iCloud, so it moves easily to another provider. Scope, also 19 Oct: **one FreeCAD at a time**; simultaneous edits from several instances are not part of this step.

Layout. A sync service makes conflict copies only when *the same file* changes in two places, or changes again while an earlier version is still uploading. The vault is laid out so that never happens:

Path in the vault folder	Rule
`<name>.bwvault`	The **ref file**. Identifies the vault (name, id, format version). Written once. Today `bw-hooks vault status -vault <file>` opens it; the FreeCAD panel (`PDM-04`) will open it from File → Open or a double-click.
`objects/ab/<sha256>.FCStd`	Every checked-in file version, named by its content hash. **Written once, never changed**; written under a `.tmp` name (which iCloud does not sync) and renamed when complete; checked against its hash when read.
`journal/<user>@<mac>.jsonl`	Every action (create, check out, check in, submit, approve, ECR, ECO, BOM) as one line. **Append-only, and only that user on that Mac writes it.** The state is rebuilt by replaying the lines through the reference model (`PDM-02`), so the lifecycle rules are identical.
working folder (outside the vault)	Where FreeCAD opens and saves while you work. Check-in copies the file into `objects/`; check-out copies the latest one out. Saving every few minutes never touches the vault.

If a sync service makes a copy anyway (iCloud `x 2.jsonl`, Dropbox `x (conflicted copy).jsonl`, Syncthing `x.sync-conflict-...`), the reader reads *every* journal file and keeps each event once, so a copy neither loses nor doubles anything. A half-synced last line is skipped until the rest arrives; a journal line whose file has not downloaded yet is reported as "not synced yet" rather than an error. Moving to another provider means moving the folder.

Built and tested headless. `addons/BruceWorx/bruceworx/pdm_folder.py` with the command line `bw-hooks vault` (actions `init`, `status`, `create`, `checkout`, `checkin`, `undo-checkout`, `submit`, `approve`, `reject`, `fetch`). The 12 lifecycle and BOM contract tests of `PDM-02` pass unchanged against a real folder, plus 17 folder tests: reopening rebuilds identical state; a refused action writes nothing; versions are never rewritten; the released revision freezes its file; damaged files are caught; three kinds of duplicate journal, an emptied journal, a half-written line, a missing download, and a duplicated ref file. The OpenAPI file stays as the list of operations; the folder vault implements them locally, and a server can implement them later.

iCloud: what we found

Your Documents folder is in iCloud. Desktop & Documents sync is on (`~/Library/Mobile Documents/com~apple~CloudDocs/Documents` points at `~/Documents`), so this repository is synced.

Where the duplicates are. 1 488 numbered copies, *all* in machine-written trees: build output (`build/release/share`, `src`) and 21 inside `.git` (`HEAD 2`, `index 2`, `config 2`). None among the hand-edited sources. They arrived in bursts at clone and build times (937 in one minute on 15 Oct). The copies hold older contents: iCloud kept the version it had while the build was replacing thousands of files.

Single-file saves did not reproduce it. On 19 Oct, 120 rapid rewrites of three files (save by rename, delete and recreate, overwrite in place) and three write passes over 3 000 small files on this Mac produced no copies. The trigger is heavy, fast, machine-made churn, which is exactly what the vault never does.

Excluding a folder works on this Mac (macOS 27.2). A folder whose name ends in `.nosync`, or one carrying the attribute `com.apple.fileprovider.ignore#P`, is reported by `fileproviderctl` as `isExcludedFromSync = 1` and is not uploaded. Both are undocumented by Apple and could change; the `.nosync` name is the more widely used.

Storage optimisation. With "Optimise Mac Storage" on, iCloud can evict files to save space; opening one waits for a download and fails offline. For the vault folder, turn it off or use Finder's *Keep Downloaded* on the folder.

Sharing needs a folder in iCloud Drive itself. Apple does not share the Desktop or Documents folders themselves; a shared folder has to sit in iCloud Drive. Folder sharing needs macOS 10.15.4, iOS 13.4 or iCloud for Windows 11.1 or later.

Recommended setup (yours to apply).

1. Put the vault in **iCloud Drive, not Documents**, e.g. `iCloud Drive/BruceWorx Vault/`. A vault in Documents works for you alone, but a vault in iCloud Drive can be shared later without moving it.
2. Keep the working folder out of sync: `~/BruceWorx Work` (outside Documents), or a `Work.nosync` folder.
3. Turn on *Keep Downloaded* for the vault folder.
4. This repository: move it out of Documents (for example to `~/Developer`), or at least exclude `build` and `.pixi`. Git inside iCloud is a known way to damage a repository, and its `.git` already holds 21 duplicate files. Not changed here; it is your machine and your call.

The repository, looked at again (19 Oct). `git fsck` is clean, but the duplicates show the risk. `.git/HEAD 2` reads `ref: refs/heads/.invalid` and `.git/config 2` has lost the branch and submodule settings: both are the half-written state from the clone. Had iCloud settled those conflicts the other way, the repository would have lost track of its branch and of GitHub. Excluding only `build` and `.pixi` would leave this in place, and a branch switch or rebase rewrites thousands of source files at once, which is the same burst that made the copies.

Recommended. Repository in `~/Developer`, outside iCloud. Its off-machine copy is GitHub (a pushed work branch) plus Time Machine, both of which understand git. The vault, the one thing that needs file sync, goes in `iCloud Drive/BruceWorx Vault/`, which is built for it.

Not recommended. Keeping the repository in Documents with `.git`, `build` and `.pixi` excluded. The sources would still sync, but iCloud's copy would be sources without history, and the exclusion relies on undocumented behaviour.

Moving needs a fresh `pixi install` and a full build, because both store absolute paths. Uncommitted work should be pushed to a branch first, so that it exists somewhere other than this Mac.

Limits of this step. One writer at a time. Journals are per user and Mac, so a second Mac or a second person can be added later, but two people editing at the same moment are not reconciled, and the check-out lock is a record in the journal, not a server lock. When a team needs that, the same operations move behind a server (option A below) without changing the files.

7.2 BW-PDM-01 — Evaluation memo (19 Oct 2026, before the decision)

Status. Kept for the record. Option C was scored as a *plain* shared folder with FreeCAD files saved straight into it; the decision above is a folder vault built to avoid exactly the weaknesses scored here.

Evidence	A. Ops is the PLM	B. Ops + Git LFS	C. Vault on a shared folder	D. Open-source PLM
Platform today	Ops = Next.js 16 on Railway (`bruceworks`); task API and auth are being built on branches (staging `/api/tasks` answers 401)	git-lfs 3.8 installed here; GitHub hosts the repos already	Folder on NAS or cloud drive	DocDokuPLM (Java), LibrePLM (Odoo; Windows-only connector), FreePDM (concept, no releases)
File storage cost	Railway bucket: $0.015/GB-month, S3 operations and bucket egress free	GitHub Free/Pro: 10 GiB storage + 10 GiB bandwidth per month, then $0.07/GiB storage, $0.0875/GiB bandwidth; Team: 250 GiB each	Disk you own	Self-hosting + upkeep
Locking	Server-side lease per item (contract: `/checkout`, `/checkin`)	`git lfs lock`: others cannot push a locked file	None real. **This repo already shows the failure:** iCloud left conflict copies (`BruceWorx 2.py`, `build 2.ninja`) instead of refusing	Built in
Lifecycle and ECO	Exactly the reference model (`bruceworx/pdm.py`)	In Ops (same model); Git only versions files	Must be built in the FreeCAD client	Configurable; fit with Y14.35 + ECR/ECO unproven
Multi-file assembly links	Client maps vault paths to a local workspace	Repo-relative paths work as they are	Share-relative paths	Needs a FreeCAD connector (none exists)
BOM and where-used	Ops (graph queries)	Ops	Client index	Built in
Agents / MCP	One API for FreeCAD, agents and the website	Git + Ops API	Weak	Varies
Main risk	Effort: we own a vault	Two systems to keep in step; CAD users must use Git (or a client that hides it)	Silent duplicates, no server truth	Connector effort, maintenance, Windows bias

Criterion (proposed weight)	A	B	C	D
Data integrity and locking (25 %)	5	4	2	4
Lifecycle fit, Y14.35 + ECR/ECO (20 %)	5	3	3	3
Multi-file references (15 %)	4	5	4	2
BOM and where-used (15 %)	5	4	2	5
Build effort, higher is less work (10 %)	2	3	3	2
Agent / API access (10 %)	5	4	2	3
Running cost (5 %)	5	3	5	4
Weighted score	**4.55**	**3.80**	**2.75**	**3.35**

Recommendation at the time. Prototype **A** and **B** on `LE-02001`, then decide. Rule out **C**: its failure mode (silent duplicates) is already visible in this repository. Keep **D** as a fallback only, since none has a FreeCAD connector. Either A or B uses the same reference model and contract (`PDM-02`), so no work is lost whichever wins.

Prototype plan (not pursued after the decision).

A. A small service implementing `docs/pdm-openapi.yaml` over `bruceworx/pdm.py`, with files in a Railway bucket and SHA-256 content addressing. Run the contract tests against it, then check in the `LE-02001` tree and walk one ECO. Creating the bucket and service is an outward-facing change on your Railway account, so it waits for you.

B. A private GitHub repository with LFS tracking for `*.FCStd` and locking turned on. Two clones play two engineers; measure lock conflicts and the clone size after ten revisions of the pedestal.

7.3 Data model

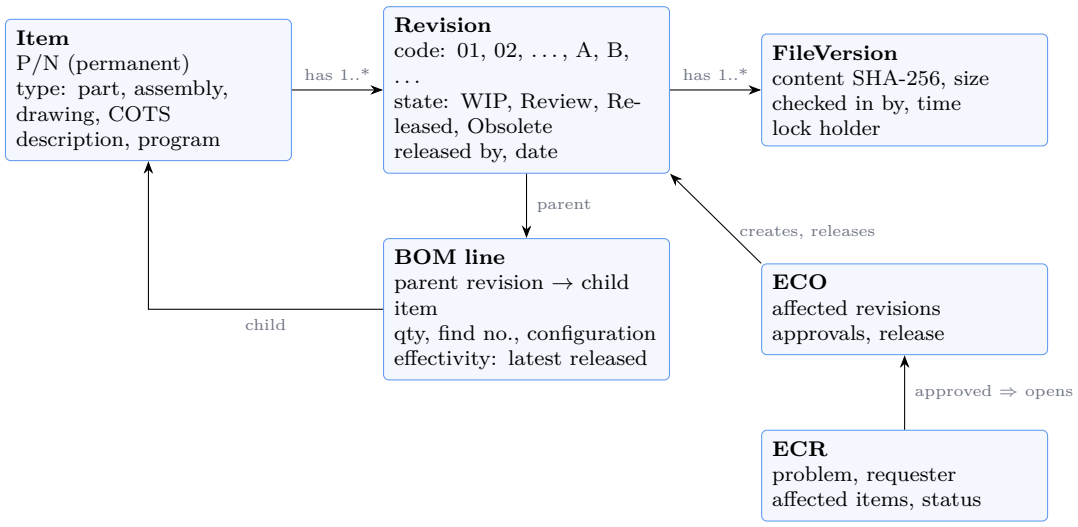


Figure 14: PDM data model. The P/N belongs to the *Item* and never changes. Revisions carry the state. File versions are content-addressed. BOM lines point from a parent *revision* to a child *item*, so a released assembly picks up the child's latest released revision unless it is pinned.

`BW-PDM-02` Data model & lifecycle spec — M2 — In progress

Specify Figure 14 and the lifecycle in Figure 15 as an API contract (OpenAPI) that any storage option implements. A drawing is its own *Item*, numbered like its part, whose revision moves *with* the part's: one ECO releases both. **Drafted 19 Oct 2026:** reference model `addons/BruceWorx/bruceworx/pdm.py` (revision sequences, lifecycle, ECR/ECO, drawing coupling, BOM explosion, where-used, cycle rejection) with 18 contract tests in `tests/test_pdm.py`, and the API contract `addons/BruceWorx/docs/pdm-openapi.yaml` (17 endpoints, including check-in/out and locks). *Acceptance:* spec approved; every transition in Figure 15 has an endpoint and a test.

Ideas absorbed: (new)

7.4 Lifecycle and revisions

Theory: revision sequences. Released revisions use letters with I, O, Q, S, X and Z skipped, as ASME Y14.35 recommends, because they read as digits. Writing the 20 usable letters as an alphabet Σ, the sequence is base-20 with no zero digit: A, B, . . . , Y, AA, AB, ... The successor function increments the last letter and carries. Pre-release revisions are zero-padded integers. The drawing prints `P/N-REV`, for example `LE-01-01001-001-02` or `LE-01-01001-001-A`, while the P/N stored on the item never includes the revision (decided).

Theory: why pessimistic locking. `.FCStd` files are zip archives of XML and B-rep data. Two concurrent edits cannot be merged, so optimistic concurrency, where both editors save and a merge

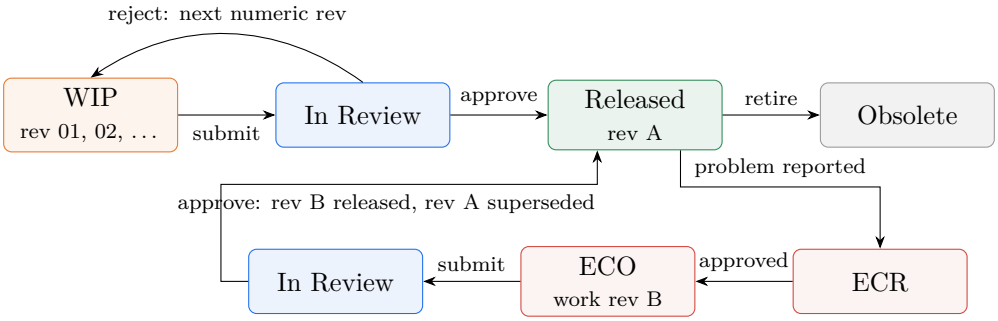


Figure 15: Lifecycle. Pre-release revisions are numeric (ASME Y14.35 practice) and every review rejection increments them. The first approval releases rev A. After that, letters change only through an approved ECR → ECO, and released files are read-only. "Superseded" is a flag on the old letter revision, not a state.

resolves it, would always end with one person's work discarded. With 2–5 editors, *pessimistic* locking (check-out grants an exclusive lock; check-in releases it) costs little waiting and avoids lost work. Locks carry a lease, and an administrator can break a stale one. A file version is identified by its SHA-256, so an unchanged check-in is a no-op and duplicate content is stored once.

`BW-PDM-03` Numbering service — M2 — In progress

The local-JSON numbering source and macro are merged (#5). The format is `{PROG}-{DT}-{PPPPP}-{SSS}-{REV}`. Under the decision, `{REV}` moves out of the P/N into revision metadata, and label and file names drop it. Then the Ops API reader (stubbed as `OpsApiNumberSource`) allocates numbers centrally, with a bearer token from `BW_OPS_TOKEN`. *Acceptance:* P/Ns come from Ops; existing LE numbers migrate with only the revision suffix removed.

Ideas absorbed: FCB-P2-03/04/05

`BW-PDM-04` FreeCAD vault client — M2 — To do

A Vault panel and tree overlays: *checked in*, *checked out by me*, *locked by someone else*, *newer version available*. Commands: check out, check in (with comment), undo check-out, get latest, and get a specific revision. Opening a locked file opens it read-only. Saving a file you have not checked out offers to check it out first. Assembly links resolve through the vault, so `LE-02001` opens identically on every machine. Works over the folder vault (§7.1); the command-line `vault` commands already drive it. *Acceptance:* a check-out/check-in round trip of `LE-02001` from an iCloud Drive vault inside FreeCAD; the status icons are always correct. Locking between several users is a later step.

Ideas absorbed: (new)

`BW-PDM-05` Lifecycle & change control — M2 — To do

Implements Figure 15, with approvals in Ops. Release freezes files read-only, publishes drawings (`DRW-08`), and builds the release package (`MFG-04`). The ECR captures the problem; the ECO groups affected items, opens their next letter revision in WIP, and releases them together. *Acceptance:* rev A → ECR → ECO → rev B round trip on `LE-01001` and its drawing.

Ideas absorbed: (new)

BW-PDM-06 BOM sync & where-used M2 To do

On check-in of an assembly, push its BOM (**ASM-04**) to Ops as BOM lines. Where-used is the reverse traversal from §5. Released assemblies resolve children to the latest *released* revision. A WIP assembly may also use WIP children, and is flagged when it does. *Acceptance:* where-used of a bolt lists every assembly revision that uses it.

Ideas absorbed: (new)

BW-PDM-07 Pilot: LE-02001 & CAD register M2 To do

The first real release: build the pedestal tree (`LE-01001` → `LE-01010` → `LE-01009` → `LE-01011` → `LE-01002`), link the CAD register in the BruceWorx app, run the design-freeze checklist and validation macro, and release rev A with drawings and package. *Acceptance:* `LE-02001` rev A released entirely from FreeCAD.

Ideas absorbed: FCB-P2-15/16/17/18, P2-21

BW-PDM-08 Migration from Fusion M6 To do

Bulk-import Fusion exports (STEP, plus drawings as PDF) into the vault with P/N and revision mapping, re-establish assembly structure, and record provenance. The FC-T01 switch criteria become the checklist for M6. *Acceptance:* every LE part imported, numbered and released; FC-T01 criteria met.

Ideas absorbed: FC-T01, FCB-P2-22

8 Track SIM — Simulation

The solvers are already in the box. CalculiX in FreeCAD exposes `static`, `frequency`, `thermomech`, `check` and `buckling` (`femsolver/calculix/solver.py:44`), with contact in 1.1, and Elmer adds thermal and multiphysics. What is missing compared with Fusion's simulation workspace is the *workflow*: study templates, sensible defaults, a safety-factor result, FDM-aware materials, and verified reports. A grep of `src/Mod/Fem` finds no safety-factor result and no orthotropic material model. The material library has ABS, PLA, PA6 and PET, but no PETG or carbon-filled nylon. Scope as decided: static + contact, modal, buckling, thermal and thermal stress now; CFD later.

BW-SIM-01 Study setup (Fusion-style) M4 To do

A Simulation workspace with study types *Static stress*, *Modal frequencies*, *Structural buckling*, *Thermal* and *Thermal stress*. Each is a template that creates the FEM analysis, mesh (with curvature-based refinement), material assignment from the part, and the loads and constraints panel, with a pre-check that lists missing inputs before solving. *Acceptance:* every study type runs from the template on a corpus part, with no FEM-workbench steps.

Ideas absorbed: (new)

BW-SIM-02 Contacts, bolts & preload M4 To do

Automatic contact detection between touching assembly faces (bonded by default, separation where chosen), and **bolt connectors** with preload that replace modelled fasteners in the mesh. *Acceptance:* a bolted-bracket benchmark is within 5 % of the reference solution.

Ideas absorbed: (new)

BW-SIM-03 Materials incl. FDM — M4 — To do

Library entries for PETG, ABS, PLA, PA-CF, Al 6061-T6 and steel, with yield, ultimate, modulus, Poisson's ratio, density, thermal conductivity, expansion and heat-deflection temperature. Printed materials get a **knockdown** model by default and an optional **orthotropic** model oriented by the part's print direction (shared with `MFG-01`). The user picks per study. *Acceptance:* the same bracket printed in two orientations shows a lower safety factor in the Z-weak one.

Ideas absorbed: (new)

Theory: FDM material models. A printed part is stronger and stiffer along the deposited roads (the XY plane) than across layers (Z), where strength depends on interlayer bonding.

1. **Isotropic with knockdown.** Use isotropic E, ν, and an allowable stress $\sigma_{\text{allow}} = \eta\,\sigma_y$ with $\eta \in (0, 1]$ per material and process, for example 0.5–0.7 for PETG with Z loading possible. It is conservative, simple and easy to validate with one coupon test per material.
2. **Orthotropic (transversely isotropic) by orientation.** With in-plane axes 1, 2 (XY, isotropic) and the build axis 3, the compliance relation $\boldsymbol{\varepsilon} = S\boldsymbol{\sigma}$ uses

$$S = \begin{pmatrix} \frac{1}{E_p} & -\frac{\nu_p}{E_p} & -\frac{\nu_{zp}}{E_z} & & & \\ -\frac{\nu_p}{E_p} & \frac{1}{E_p} & -\frac{\nu_{zp}}{E_z} & & & \\ -\frac{\nu_{pz}}{E_p} & -\frac{\nu_{pz}}{E_p} & \frac{1}{E_z} & & & \\ & & & \frac{1}{G_{zp}} & & \\ & & & & \frac{1}{G_{zp}} & \\ & & & & & \frac{2(1+\nu_p)}{E_p} \end{pmatrix}, \qquad \frac{\nu_{pz}}{E_p} = \frac{\nu_{zp}}{E_z},$$

   so there are 5 independent constants $(E_p, E_z, \nu_p, \nu_{zp}, G_{zp})$. The material frame is rotated into the part frame from the print-direction property. Failure uses a Tsai–Hill-type index with separate in-plane and interlayer strengths (X_p, Z), so a Z-loaded design fails earlier, as it does on a printer.

CalculiX supports anisotropic elastic materials through its `*ELASTIC, TYPE=ENGINEERING CONSTANTS` card [verify]. The work is in the FreeCAD material model, the orientation transfer and the result evaluation.

BW-SIM-04 Results & reports — M4 — To do

Results: von Mises, principal stresses, displacement, **safety factor** (minimum and field), mode shapes with frequencies, buckling load factors, and temperatures. Probes, a **mesh-convergence study** (automatic refinement until the quantity of interest changes by less than a tolerance), and a PDF report generated from the study. *Acceptance:* the report regenerates from the study and includes the convergence plot.

Ideas absorbed: (new)

Theory: what each study solves. After discretisation, with stiffness K, mass M, geometric (stress) stiffness K_σ, conductivity K_T and heat load $\mathbf{q}$:

$$\text{static:}\quad K\mathbf{u} = \mathbf{f}, \qquad \sigma_{vM} = \sqrt{\tfrac{1}{2}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right]}, \qquad SF = \frac{\sigma_{\text{allow}}}{\max \sigma_{vM}}$$

$$\text{modal:}\quad K\boldsymbol{\phi} = \omega^2 M\boldsymbol{\phi}, \qquad f_i = \omega_i/2\pi$$

$$\text{buckling:}\quad (K + \lambda K_\sigma)\boldsymbol{\phi} = \mathbf{0}, \qquad P_{\text{cr}} = \lambda_1 P_{\text{applied}}$$

$$\text{thermal:}\quad K_T\mathbf{T} = \mathbf{q}; \quad \text{thermal stress: } K\mathbf{u} = \mathbf{f} + \mathbf{f}_{\text{th}}(\mathbf{T} - \mathbf{T}_0)$$

For drone arms, the modal study matters because the first bending frequency must stay away from motor rotation frequencies, $f_1 \notin [0.8, 1.2] \cdot \text{RPM}/60$. **Convergence:** for element size h and

order p, the error in a smooth quantity decreases as $O(h^{p+1})$. Three meshes refined by a ratio r give an extrapolated value (Richardson), $Q_\infty \approx Q_3 + (Q_3 - Q_2)/(r^{p+1} - 1)$. The report shows $|Q_3 - Q_\infty|/|Q_\infty|$ as the discretisation error.

`BW-SIM-05` Verification benchmarks — M4 — To do

Closed-form checks run in CI on every solver or FreeCAD update: cantilever tip deflection $\delta = PL^3/3EI$; plate with a small central hole in tension, $K_t \to 3$; Euler column $P_{\text{cr}} = \pi^2 EI/(KL)^2$; cantilever first mode $\omega_1 = 1.875^2\sqrt{EI/(\rho AL^4)}$; plane wall conduction $q = k\,\Delta T/t$. *Acceptance:* all benchmarks within the stated tolerance (2–5 %) in CI.

Ideas absorbed: (new)

`BW-SIM-06` CFD / airflow (later) — M5 — To do

Deferred by decision. When it starts, the candidate is the CfdOF add-on driving OpenFOAM, for prop wash on arms and cooling airflow through electronics enclosures. Scope is set at that time.

Ideas absorbed: (new)

9 Track MFG — Manufacturing outputs

BruceWorx parts go to three processes: FDM printing (PETG/ABS), CNC machining and sheet metal / laser. FreeCAD already exports 3MF (Mesh workbench) and DXF (Draft importers), and its CAM workbench ships post-processors, including GRBL and LinuxCNC. This track connects those to the model and to the release.

`BW-MFG-01` 3D-print prep — M3 — To do

- A **print orientation** property on each part (build direction plus a rotation about it). It is shown as a build-plate gizmo and used by the overhang map (`MOD-11`) and the orthotropic material (`SIM-03`).
- **3MF export** with orientation applied, units, part name `P/N-REV`, material and colour metadata.
- **Slicer handoff**: open the 3MF directly in OrcaSlicer, Bambu Studio or PrusaSlicer, configurable, with a per-material profile hint.
- **Insert and print-hole presets**: heat-set insert holes by insert size, plus clearance and horizontal-hole teardrop shapes (from `MOD-05`).

Acceptance: a part opens in the slicer already oriented, with its name and material set.

Ideas absorbed: (new)

Theory: choosing an orientation. For a candidate build direction $\mathbf{b}$, score

$$J(\mathbf{b}) = w_1 \sum_{f:\ \mathbf{n}_f \cdot (-\mathbf{b}) > \cos\theta_c} A_f \;+\; w_2\, h(\mathbf{b}) \;+\; w_3\, \left|\boldsymbol{\sigma}_{\text{dominant}} \cdot \mathbf{b}\right|,$$

which adds supported area, build height and the alignment of the main load with the weak Z axis (from a static study, if one exists). Evaluating the faces of the convex hull plus the principal axes as candidates, and taking the minimum, gives a suggested orientation that the user can accept or override.

BW-MFG-02 CNC CAM — M3 — To do

Use FreeCAD CAM with BruceWorx defaults: a **setup template** (stock from the bounding box plus allowance, work offset at a datum corner), a **tool library** matching the shop's tools, operation templates (face, adaptive clearing, profile, drill and chamfer), and a **post-processor** for the target machine (open question). *Acceptance:* a corpus part posts G-code that simulates clean in the CAM simulator.

Ideas absorbed: (new)

BW-MFG-03 Sheet-metal flat DXF — M3 — To do

Export each sheet-metal part's flat pattern (`MOD-06`) as DXF, with layers `CUT` (outline and holes), `BEND_UP`/`BEND_DOWN` (bend lines) and `ETCH` (part number), and a bend note block. File name `P/N-REV.dxf`. Units are mm, and splines are converted to arcs where the shop's CAM requires it. *Acceptance:* the laser shop accepts the DXF without edits.

Ideas absorbed: (new)

BW-MFG-04 Release package — M2 — To do

When a revision is released (`PDM-05`), build a package: STEP AP242 per part and assembly, the PDF drawing, DXF flats, 3MF prints and a CSV BOM, plus a manifest with SHA-256 checksums. The package is attached to the revision in the vault. *Acceptance:* rebuilding the package from the released files gives identical checksums for every deterministic output.

Ideas absorbed: (new)

Note on determinism. STEP and PDF writers embed timestamps. The package normalises them, for example by setting the STEP header time to the release time, so the checksum identifies *content*. A vendor who receives `LE-02001-A` can then verify that the files are exactly the released ones.

10 Track ROB — Robotics and drone kits

These epics turn the general tracks into BruceWorx's own product domain: the Leonardo robot, servo arms and quadcopters. Each one builds on a general epic, so nothing here forks the core design.

BW-ROB-01 COTS library & insert — M1 — To do

A COTS library folder with STEP models or datasheet envelopes plus per-part metadata: mass, CG offset, mount pattern, category, supplier P/N, and for motors, kv and maximum thrust. The first set: NEMA17, 2207/2306 brushless motors, SG90/MG996R servos, 4-in-1 ESCs (30.5/20 mm), 5" props (disk envelope), and 608/6800 bearings (parametric). The insert command has a category picker and places **mounting LCSs**, so `ASM-03` quick joints and `ASM-09` fasteners work immediately. COTS items get COTS P/Ns in PDM. *Acceptance:* a NEMA17 and a 2207 motor are inserted and jointed by their mounting LCS.

Ideas absorbed: FCB-P2-09/10, P2R-01..04

BW-ROB-02 Frame & arm templates — M3 — To do

A parametric quad X-frame (wheelbase, arm width, motor mount pattern) and a servo link/bracket (link length, servo size), built as **configurations** (`MOD-03`) rather than separate spreadsheets. *Acceptance:* the template regenerates cleanly for three wheelbases.

Ideas absorbed: FCB-P2R-05/06

BW-ROB-03 Mass, CG & inertia report — M1 — To do

Per-part mass from density or a COTS override; totals, CG and the inertia tensor; CSV export and a 3D CG marker. It builds on `MOD-11`. *Acceptance:* the X-frame example in Figure 16: 388.1 g, CG at $(2.45, 0, -2.73)$ mm.

Ideas absorbed: FCB-P2R-07

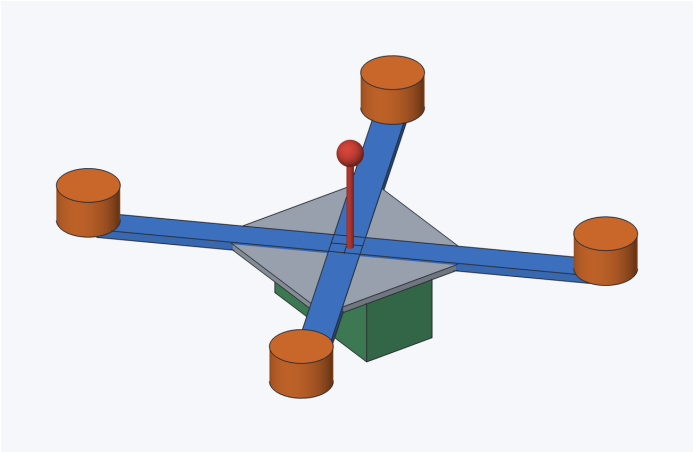


Figure 16: Quad X-frame: carbon plate and arms from density (1.6 g/cm^3), four 32 g motors and a 190 g battery as COTS overrides. The battery is mounted 5 mm forward, which puts the CG (red plumb line and marker) at $x = 2.45$ mm. That is exactly what the balance check in `ROB-04` must flag. Computed in headless FreeCAD.

BW-ROB-04 Drone balance & thrust-to-weight — M3 — To do

Compare the CG with the centroid of the motor axes, and report thrust-to-weight $T/W = \sum_i T_{\max,i}/(mg)$ from COTS data. Warn when the horizontal CG offset exceeds a fraction of the arm length, or $T/W < 2$. *Acceptance:* the check flags the offset battery in Figure 16.

Ideas absorbed: FCB-P2R-08

BW-ROB-05 Kinematic chains & joint drive — M5 — To do

Build a link chain from revolute joints, show a joint table, read out end-effector forward kinematics, drive the chain from the table through the stock Simulation (servo-sweep presets), and export video. *Acceptance:* the 2-DOF arm's FK matches a hand calculation.

Ideas absorbed: FCB-P2A-09/10

Theory: forward kinematics. Each joint i contributes a homogeneous transform $T_i(\theta_i) = \begin{pmatrix} R(\mathbf{a}_i, \theta_i) & \mathbf{p}_i - R(\mathbf{a}_i, \theta_i)\mathbf{p}_i \\ \mathbf{0}^\top & 1 \end{pmatrix}$, a rotation about its axis through $\mathbf{p}_i$ in the reference pose. The end-effector pose is the product

$$T_{\mathrm{ee}}(\boldsymbol{\theta}) = T_1(\theta_1)\, T_2(\theta_2) \cdots T_n(\theta_n)\, T_{\mathrm{ee}}(\mathbf{0}),$$

the product-of-exponentials form, which needs no Denavit–Hartenberg frames and can be read directly from the assembly's joints. For the planar 2-DOF arm, this reduces to $x = \ell_1 \cos\theta_1 + \ell_2 \cos(\theta_1 + \theta_2)$ and $y = \ell_1 \sin\theta_1 + \ell_2 \sin(\theta_1 + \theta_2)$, the acceptance check.

11 Track ECD — ECAD bridge (KiCad)

BW-ECD-01 ECAD intake & adopt-vs-build — M1 — To do

Requirements intake from the ECAD owner (KiCad version, outline and keep-outs, direction of placement sync, STEP needs, sample board). Evaluate the KiCad StepUp add-on on current FreeCAD with a KiCad 9 board, including its AGPL licence and maintenance. Then a decision memo: adopt StepUp unmodified, or build a bridge on `kicad-cli` and the KiCad 9 IPC API. The Ops sample data shows this gate as done, but the result isn't recorded (open question).

Ideas absorbed: FCB-P2E-01..04

BW-ECD-02 Board STEP into assembly — M3 — To do

Export with `kicad-cli pcb export step`, and insert into the assembly with an LCS at the board origin, so the board joints like any component.

Ideas absorbed: FCB-P2E-05

BW-ECD-03 Board outline round-trip — M3 — To do

Pull `Edge.Cuts` into a Sketcher sketch and push edits back to `.kicad_pcb`. *Acceptance:* the round-trip outline delta is below 0.01 mm on the sample board.

Ideas absorbed: FCB-P2E-06, P2E-09

BW-ECD-04 Placement sync & enclosure fit — M3 — To do

Push connector and mounting-hole positions from MCAD to ECAD and back. Check board against housing with the interference command (**ASM-06**). *Acceptance:* the fit check flags a connector moved into a wall.

Ideas absorbed: FCB-P2E-07/08

12 Track AUT — AI and automation

BW-AUT-01 Headless hooks & CLI — M0 — Done

Merged in #5: `addons/BruceWorx/hooks.py` exports STEP, lists the document tree as JSON, and numbers and renames parts through FreeCADCmd. Every later automation epic builds on it.

Ideas absorbed: FCB-P2-11..14

BW-AUT-02 bruceworx.api — M1 — To do

A stable Python module over the hooks: create and edit parts, set parameters (**MOD-02**), read the tree, export STEP/STL/3MF, run inspections (**MOD-11**). One undo transaction per call. *Acceptance:* the API is covered by headless tests in CI.

Ideas absorbed: FCB-P2AI-03

BW-AUT-03 MCP server — M1 — To do

Choose a base among the MIT and LGPL FreeCAD MCP servers (spike plus gate, including whether raw `execute_code` is allowed), then add BruceWorx tools: tree, create/edit, export, numbering, COTS insert, mass report. Includes a headless mode on FreeCADCmd for cloud agents, and security: localhost by default, a token, and an audit log. Configuration docs for the agents in use, and integration tests in CI. *Acceptance:* an agent creates, queries and exports a part headless in CI.

Ideas absorbed: FCB-P2AI-01/02, 04..08

BW-AUT-04 Assistant panel — M5 — To do

A dock with chat, a tool-call preview, and apply/undo per action. It calls `bruceworx.api` and MCP, and gets a UX mock and design gate first. *Acceptance:* a try-out on real CAD work is signed off.

Ideas absorbed: FCB-P2AI-09..15

BW-AUT-05 Command line — M5 — To do

A bottom command line accepting AutoCAD-style aliases, inline Python, CadQuery/build123d scripts and OpenSCAD, plus a `CodeFeature` that keeps a script as a parametric, trust-gated feature. *Acceptance:* `CPATTERN @Bolt Q 8` reproduces Figure 4d, and a build123d box lands as a solid in the timeline.

Ideas absorbed: (new, from the UI plan)

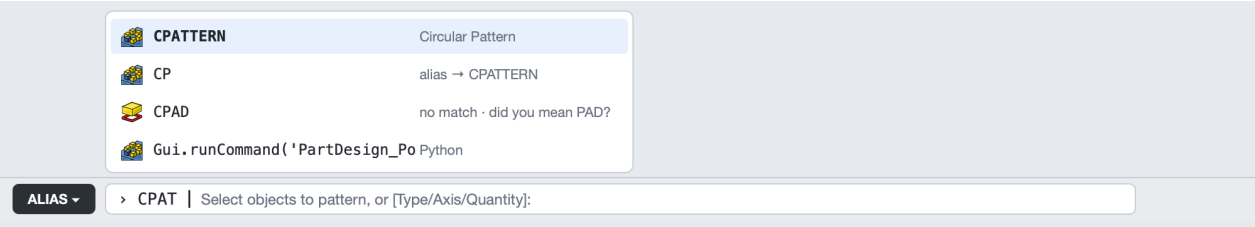


Figure 17: Command line typing `CPAT` (working mock): ranked completions, the alias expansion, a "did you mean" suggestion and the Python equivalent, with an AutoCAD-style `[Option/Option]` prompt.

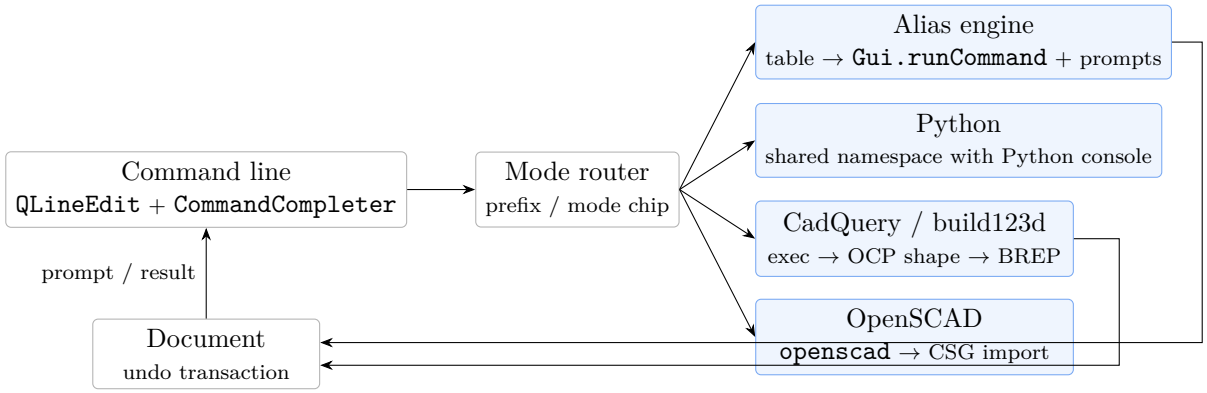


Figure 18: Command-line pipeline; each run is one undo transaction.

Design. A bottom toolbar spanning the window, with focus key `:` or `Ctrl+‘`. An explicit mode chip (`ALIAS/PY/CQ/B3D/SCAD`) plus prefixes (`»`, `cq:`, `b3d:`, `scad:`), with alias mode as the default. Completion reuses FreeCAD's `Gui::CommandCompleter`. Python runs in the console interpreter's namespace. OpenSCAD uses the existing importer. No maintained add-on provides an AutoCAD-style line, and none bridges build123d into FreeCAD (only the reverse, `ocp-freecad-cam`), so both are written here.

Theory: alias grammar and prompts. Aliases live in a plain file in the spirit of AutoCAD's `acad.pgp`. A line is parsed as

```
line     := command ( WS arg )*
command  := ALIAS | COMMAND_NAME            ; case-insensitive
arg      := number unit? | option | ref
option   := IDENT                           ; matched against [A/B/C] prompt options
ref      := '@' object-label ( '.' subelement )?   ; e.g. @Flange.Face3
```

Missing arguments are asked for by a prompt state machine, SELECT → OPTION* → VALUE* → COMMIT. Viewport selection, a typed value or Enter advances it, and Esc aborts the transaction.

Theory: ranking completions. For query q and candidate c,

$$\sigma(q,c) = 4\,[\,q \sqsubseteq_{\text{prefix}} c\,] + 2\,[\,q \sqsubseteq_{\text{initials}} c\,] + \frac{|q|}{\text{span}(q,c)} + \lambda \log(1+f_c),$$

where span is the shortest window of c containing q as a subsequence and f_c is the user's usage count. Unknown input gets the smallest Damerau–Levenshtein candidate ("did you mean PAD?").

Theory: code-CAD bridge. CadQuery and build123d wrap OCCT through OCP (pybind11), and FreeCAD through `TopoShape`. Even when both load the same OCCT libraries, passing raw `TopoDS_Shape` pointers would couple two binding ABIs. The bridge therefore serialises through OCCT's BREP format: `OCP shape` $\xrightarrow{\texttt{BRepTools::Write}}$ BREP $\xrightarrow{\texttt{importBrepFromString}}$ `TopoShape`. Measured on the flange (15 faces, 10.9 kB): 0.17 ms to write and 0.12 ms to read on the FreeCAD side, with exact volume round-trip. The OCP-side write is still to be measured [verify]. `CodeFeatures` re-execute on recompute, so they start *untrusted*, with recompute disabled until the user approves them, the same trust model as FreeCAD macros.

Dependencies and the upstream-main decision.

Package (conda-forge)	Version	Needs	Fit with the current build (OCCT 7.8.1, Python 3.11)
`ocp`	7.8.1.2	occt 7.8.1 `all_h869bdd7_203`	Same OCCT build hash as `pixi.lock` (osx-arm64)
`build123d`	0.9.1	ocp ≥7.8,<7.9	OK
`cadquery`	2.7.0	ocp ≥7.8.1,<7.9	OK, pin `<2.8`
`cadquery`	2.8.0	ocp 7.9.3.1	Conflicts with OCCT 7.8
`openscad`	—	—	Not on conda-forge; separate binary

Tracking upstream main (`FND-03`) means FreeCAD's OCCT will move to 7.9 or later. The OCP pin must then move with it: CadQuery 2.8 already pairs with OCP 7.9.3. The pin is re-solved whenever the nightly rebase changes `occt` in `pixi.toml`, and the command line's code-CAD modes are disabled with a message, never broken, if no matching OCP exists yet.

13 Track UIX — UI and styling (after M2)

Styling comes after the functional tracks, as decided. Three epics are already merged or under way: the preference pack and viewport styling (#8), the macOS title bar and overlay tree (#6), and the timeline spike (#7). The rest of this chapter is the Fusion-style layout from the earlier UI plan, kept as the *starting brief* for the redesign. Its mocks will be redrawn in `UIX-09`.

`BW-UIX-01` Theme, pref pack & viewport styling — UI — **Done**

Merged (#8). Follow-ups: the anti-aliasing check on Retina, and the mechanical user's try-out and sign-off.

Ideas absorbed: FCB-P1-01..05, P1V-01..07, P1V-10..18

BW-UIX-02 macOS title bar — UI — Done

Merged (#6): unified title bar and native window buttons behind a preference.

Ideas absorbed: FCB-P1F-01..05, P1F-26

BW-UIX-03 Browser (overlay model tree) — UI — In progress

Prefs/QSS version merged (#6). The free-floating panel (any inset, free height) needs a small `OverlayManager` patch (FCB-P1F-11). Also fix it not showing on first run (**FND-06**).

Ideas absorbed: FCB-P1F-06..14, P1F-27

BW-UIX-04 History timeline — UI — In progress

Spike and mock merged (#7). Remaining: the full bar with marker drag, edit, suppress and reorder (where feasible), one-row layout, playback buttons, and a path into core.

Ideas absorbed: FCB-P1F-15..25, P1F-28

BW-UIX-05 Workspaces & tabs (Fusion naming) — UI — To do

Design / Simulation / Manufacture / Drawing workspaces; Design tabs Solid, Surface, Mesh, Sheet Metal and Utilities, plus a contextual Sketch tab. Extra workbenches are toggled in Preferences.

Ideas absorbed: (new, from the UI plan)

BW-UIX-06 Toolbar & panels — UI — To do

Captioned tool groups at one icon size, a floating command dialog, document tabs on top, and a one-row timeline.

Ideas absorbed: (new, from the UI plan)

BW-UIX-07 Task panel polish — UI — To do

Sketcher, PartDesign and TechDraw dialogs.

Ideas absorbed: FCB-P1-06/07/08

BW-UIX-08 Matcap shading — UI — To do

A matcap draw style behind a preference; with the preference off, rendering is stock.

Ideas absorbed: FCB-P1V-08/09/13/14/15

BW-UIX-09 Fresh UI design & re-render figures — UI — To do

A new visual design for the whole application, then regenerate every figure in this document with the scripts in `scripts/`.

Ideas absorbed: (new)

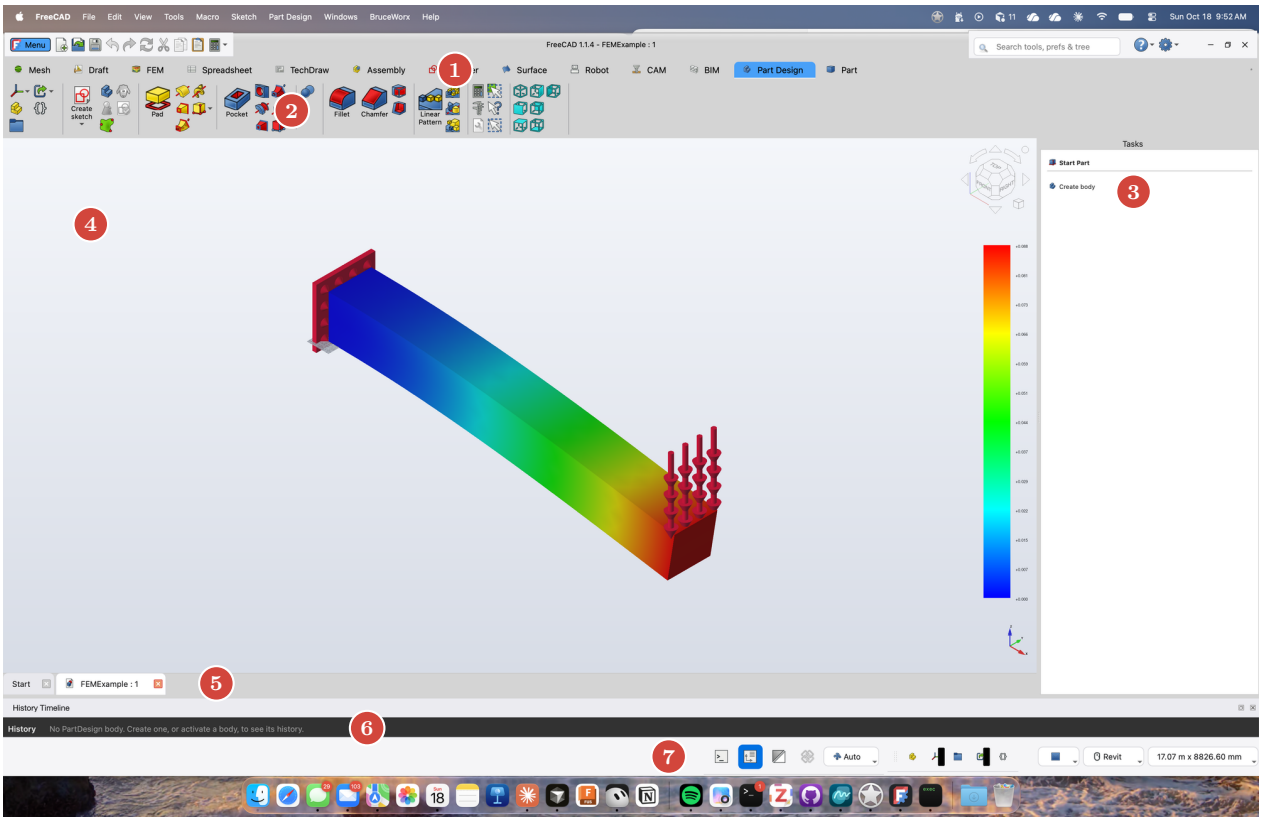


Figure 19: The window as of 18 Oct 2026, with issues numbered.

13.1 Starting brief: review of the current window

#	Issue	Cause / fix
1	13 workbench tabs	Pack trim not applied (`FND-06`); replaced by workspaces (`UIX-05`)
2	Icon grid in mixed sizes, few labels	Captioned groups, one size (`UIX-06`)
3	Full-height Tasks column	Floating command dialog (`UIX-06`)
4	No model tree	Overlay tree not shown (`FND-06`, `UIX-03`)
5	Document tabs at the bottom	Hard-coded `South` at `MainWindow.cpp:397`; make it a preference
6	Timeline has two headers	Dock title + bar header; one 40 px row (`UIX-04`)
7	Crowded status bar	Readouts move beside the command line (`AUT-05`)

13.2 Starting brief: proposed layout

Workspace	Tab	Built from
Design	Solid	PartDesign, Part booleans, Assembly (Assemble), datums (Construct), Measure (Inspect)
	Surface	Surface workbench + Part surface tools
	Mesh	Mesh and MeshPart
	Sheet Metal	SheetMetal add-on (bundled)
	Utilities	Spreadsheet, macros, Material, parameters (`MOD-02`)
	Sketch	Contextual while a sketch is open
Simulation	—	FEM, with the study templates of `SIM-01`
Manufacture	—	CAM, print prep, flats (§9)
Drawing	—	TechDraw (§6)
More…	—	Each extra workbench toggled on Preferences → BruceWorx → Workspaces

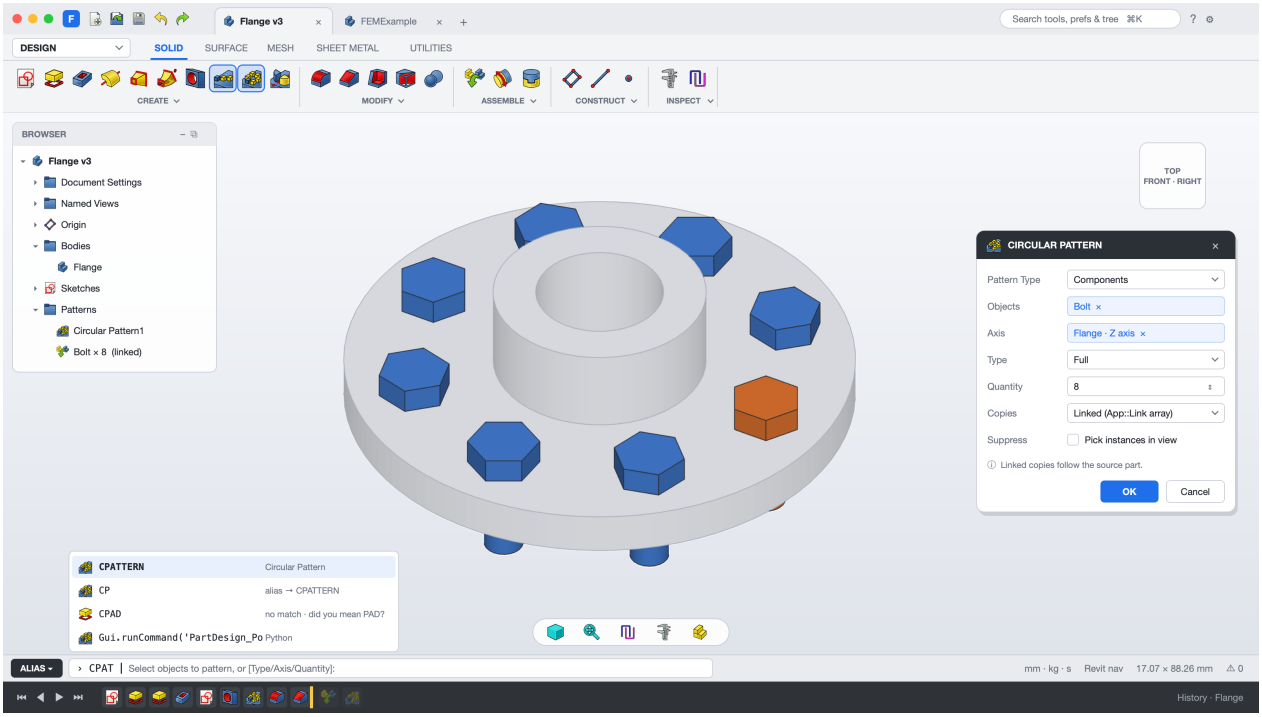


Figure 20: Working mock of the Fusion-style layout: document tabs and search in the title bar; workspace selector and tabs; captioned tool groups (Rectangular and Circular Pattern inside CREATE); Browser top-left; floating command dialog; navigation pill; command line; timeline. FreeCAD's own icons, drawn offscreen.

Selector model. A workspace w fixes a workbench $\beta(w)$, and a tab t fixes a toolbar set $\tau(w, t)$. Entering sketch edit pushes the contextual tab, and leaving pops it. Switching tabs within one β never switches workbench, so there is no toolbar rebuild or flicker. Implementation: a C++ `WorkspaceSelector` (selector type 2), a `workspaces.json` mapping, and a Python `WorkbenchManipulator` that composes tabs from several workbenches.

14 Track KRN — Kernel study (background)

A study, not a build. It answers what it would cost to put FreeCAD on a different geometry kernel, without blocking any parity work.

`BW-KRN-01` OCCT coupling survey — BG — To do

Script-count OCCT includes and call sites per module (Part, PartDesign, Sketcher, TechDraw, Surface, FEM, CAM). Analyse the topological-naming and element-map dependencies (`TopoShapeExpansion`, `ShapeMapper`, `ElementMap`) and how far OCCT leaks into Python bindings. Write option A (a facade over `TopoShape`) as a design note.

Ideas absorbed: FCB-P3-01..06

`BW-KRN-02` STEP exchange harness — BG — To do

A separate harness repository: LE corpus STEP files (shared with `FND-05`), round-trip metrics (volume, bounding box, face and edge counts, validity), a minimal kernel stub that reads and writes primitives, and Linux CI.

Ideas absorbed: FCB-P3-07..11

`BW-KRN-03` Kernel decision BG To do

An A/B/C memo with effort estimates, reviewed, then a gate. Nothing in tracks MOD–MFG waits on it. The parity work uses `TopoShape` APIs, so it carries over under option A.

Ideas absorbed: FCB-P3-12/13/14

15 Open questions

These are decisions the roadmap cannot make for you. None of them blocks M0. Each lists the epic it gates.

Q1. **FC-T01 switch criteria** (`PDM-08`, M6). The Ops data shows the gate "Bruce sets FC-T01 switch criteria" as done, but the criteria aren't recorded. What are they? They become M6's checklist and the scope filter on the parity matrix.

Q2. **Part numbers for configurations** (`MOD-03`). Proposed: each configuration that is manufactured separately gets its own P/N (§4). Or should variants share a P/N with a dash suffix?

Q3. **Drawing numbers** (`PDM-02`, `DRW-01`). Is a drawing its own item with a document-type code (`{DT}`) different from its part, or does it share the part's P/N? Their revisions move together either way.

Q4. **CNC machine and controller** (`MFG-02`). Which machine(s) and controller (GRBL, LinuxCNC, Haas, ...)? This picks the post-processor and the tool library.

Q5. **OpenSCAD binary** (`AUT-05`). Ship the official OpenSCAD app (GPL-2, as a separate executable) in the DMG, or require a user install with a hint?

Q6. **Pattern Compute: Adjust** (`MOD-01`). OK to ship *Identical* only and add *Adjust* later?

Q7. **Joints on patterned components** (`MOD-01`). Leave instances free, like Fusion, or copy the seed's joints?

Q8. **Upstreaming** (`FND-03`). With upstream main tracked continuously, should generic fixes (like `Spacings2`) and small core patches go upstream by default to shrink the conflict budget?

Q9. **Fastener library** (`MOD-05`). Adopt the FreeCAD Fasteners add-on if its licence fits, or build on the COTS schema only?

Q10. **PDM evaluation weights** (`PDM-01`). Confirm or change the proposed weights in §7.

Q11. **ECAD bridge decision** (`ECD-01`). Was "adopt StepUp vs build" decided (it shows as done in the sample data)? If so, which way?

Q12. **FDM knockdown factors** (`SIM-03`). Run coupon tests on our printers and materials, or start from published values?

Q13. **Command aliases** (`AUT-05`). Start from AutoCAD names (`EXT`, `F`, `AR`) or Fusion's single keys (`E`, `F`, `H`)?

Q14. **Distribution** (`FND-02`). Will anyone outside the team install builds before M6? If so, signing and notarisation move into M0.

A Parity matrix

● full ◑ partial ○ none [A] add-on only. Fusion = Fusion 360 with its extensions, SW = SolidWorks 2026 (Premium/PDM/Simulation), both from public vendor and reseller material (Oct 2026). A cell marked † is the vendor capability we could not confirm from that material. The

FreeCAD column is *1.1.4 as built in this repository*, with the evidence found in `src/` or by headless experiment. "Out of scope" rows were not selected in the interview and have no epic.

Capability	Fusion	SW	FC	FreeCAD evidence / note	Epic
Sketch and part					
Parametric sketch & constraints	●	●	●	Sketcher	MOD-12
Extrude, revolve, sweep, loft	●	●	●	PartDesign	—
Fillet, chamfer, shell, draft	●	●	●	PartDesign	—
Hole wizard / presets	●	●	◑	`Hole` feature; no insert presets	MOD-05
Cosmetic threads	●	●	◑	TechDraw view extension only	MOD-05
Feature patterns	●	●	◑	`LinearPattern` dir.-2 bug; no symmetric/suppress	MOD-01
Body / component patterns	●	●	◑	Draft link arrays, not assembly-aware	MOD-01
Parameters dialog	●	●	◑	Spreadsheet, `VarSet`, expressions	MOD-02
Configurations / design tables	●	●	○	none found in Spreadsheet or App	MOD-03
Multi-body, derive, skeleton	●	●	◑	`SubShapeBinder` cross-document	MOD-02
Direct edit (press-pull, move face)	●	●	◑	`Part_Defeaturing` only (verified)	MOD-04
Surfacing (patch, trim, stitch)	●	●	◑	Surface WB: Filling, Sections, Extend, Blend	MOD-08
Continuity / zebra analysis	●	●	○	—	MOD-08
Freeform / T-spline	●	◑	○	—	MOD-09
Mesh → B-rep	●	●	◑	`Part_ShapeFromMesh` (faceted), ReverseEngineering	MOD-10
Sheet metal	●	●	[A]	SheetMetal add-on (to bundle)	MOD-06
Weldments / frames	◑†	●	○	—	MOD-07
Plastic features (rib, boss, snap)	●	●	○	out of scope (no molding)	—
Measure	●	●	●	Measure WB	—
Mass properties	●	●	◑	`MeasureCOM`; no inertia report	MOD-11
Section analysis	●	●	◑	clipping / section cut	MOD-11
Draft / overhang analysis	●	●	○	—	MOD-11
Wall-thickness analysis	◑†	●	○	—	MOD-11
Assembly					
Joints / mates	●	●	●	Assembly WB (Ondsel solver)	ASM-03
Joint limits while dragging	●	●	◑	limit properties exist; drag to check	ASM-03
Rigid / flexible sub-assemblies	●	●	◑	to audit	ASM-05
Exploded views	●	●	●	`Assembly_CreateView`	ASM-04
BOM (indented, flattened, props)	●	●	◑	4 auto columns + custom	ASM-04
Interference / clearance	●	●	○	no command found	ASM-06
In-context editing	●	●	◑	`SubShapeBinder`	ASM-07
Large-assembly mode	◑	●	○	no lightweight mode found	ASM-08
Smart fasteners	◑†	●	○	—	ASM-09
Standard-parts library	●	●	[A]	Fasteners add-on	MOD-05
Kinematic motion	●	●	●	`Assembly_CreateSimulation` (1.1)	ROB-05
Dynamic motion	◑	●	○	solver dynamics not exposed	ASM-10
Simulation					
Linear static	●	●	●	CalculiX `static`	SIM-01
Contact & bolts	●	●	◑	contact (1.1); no bolt connector	SIM-02
Modal	●	●	●	CalculiX `frequency`	SIM-01
Buckling	●	●	●	CalculiX `buckling`	SIM-01
Thermal / thermal stress	●	●	●	`thermomech`, Elmer	SIM-01
Study setup UX	●	●	◑	FEM WB, manual	SIM-01

Capability	Fusion	SW	FC	FreeCAD evidence / note	Epic
Safety-factor result	●	●	○	none in `femresult`	SIM-04
Orthotropic materials	◑†	●	○	none in `src/Mod/Fem`	SIM-03
Nonlinear / event	●	●	◑	out of scope	—
Fatigue	○†	●	○	out of scope	—
CFD	◑	●	◑	Elmer flow; CfdOF add-on	SIM-06
Generative / topology	●	◑	○	out of scope	—
Drawings					
Projected, section, detail, broken	●	●	●	TechDraw	DRW-06
Dimensions incl. ordinate/chain	●	●	●	TechDraw + extensions	—
GD&T (semantic)	●	●	◑	SVG symbol library only	DRW-03
Hole & thread callouts	●	●	◑	drawing-side extension	DRW-04
Parts list + balloons	●	●	◑	balloons; spreadsheet view	DRW-02
Revision table	●	●	◑	template fields only	DRW-05
Templates & autofill	●	●	●	SVG templates, autofill	DRW-01
Auto-generate drawing	●†	●	○	—	DRW-06
Flat pattern + bend table	●	●	[A]	SheetMetal add-on	DRW-07
PDF / DXF export	●	●	●	TechDraw	DRW-08
Data management					
Version history	●	●	○	—	PDM-04
Check-in/out & locking	◑†	●	○	—	PDM-04
Part numbering	●	●	◑	BruceWorx JSON source (#5)	PDM-03
Lifecycle & change orders	●	●	○	—	PDM-05
BOM sync & where-used	●	●	○	—	PDM-06
Release package	◑	◑	○	—	MFG-04
Manufacturing and other					
CAM (2.5D–5 axis)	●	●	●	CAM WB; GRBL, LinuxCNC posts	MFG-02
3D-print prep & slicer handoff	●	◑	◑	3MF export (Mesh)	MFG-01
Flat DXF for laser	●	●	[A]	SheetMetal add-on + DXF	MFG-03
Rendering	●	●	○	out of scope (add-on exists)	—
Electronics / ECAD link	●	◑	[A]	KiCad StepUp add-on	ECD-01
Scripting / API	●	●	●	Python; headless hooks (#5)	AUT-02
Cloud collaboration	●	◑	○	decided via PDM evaluation	PDM-01
AI assistant	◑	◑	○	—	AUT-04

B Ops ideas → epics

Every idea group from the BruceWorx Ops "work" list (191 items, sample data) and the epic that absorbs it. Review and gate items ("open PR + review", "GATE: merge OK", "try-out", "sign-off") are not separate epics. They are the standard *definition of done* for every epic: spike or mock where needed, design gate, build, tests and screenshots, PR review, merge gate, and a try-out by the mechanical user on a real LE part.

Ops idea group	IDs	Absorbed by
Pixi + CI setup	FCB-P0-CI-01..06	FND-01
Build & CI	FCB-P0-01, 06, 07, 08, 10	FND-01
Branding & installer	FCB-P0-02, 03, 09, 11	FND-02
Review & release gates	FCB-P0-04, 05, 16, 17, 18	FND-02 (definition of done)

Ops idea group	IDs	Absorbed by
Docs / License / Sync	FCB-P0-12..15	FND-03 (sync policy changed), FND-04, FND-02
26.3 rebase	FCB-P1-16..21	FND-03 — replaced by tracking upstream main
Addon decision	FCB-P2-01/02	FND-03
Theme & stylesheet	FCB-P1-01..03, 09	UIX-01 (done), FND-05 (screenshot script)
Defaults & toolbars	FCB-P1-04/05	UIX-01 (done)
Task panel polish	FCB-P1-06..08	UIX-07
Review & Victor (P1)	FCB-P1-10..15	FND-07, UIX-01
Viewport prefs	FCB-P1V-01..07, 10..12	UIX-01 (done)
Matcap shading	FCB-P1V-08, 09, 13..15	UIX-08
Victor review (viewport)	FCB-P1V-16..18	UIX-01
Mac stoplights	FCB-P1F-01..05	UIX-02 (done)
Floating model tree	FCB-P1F-06..14	UIX-03
History timeline	FCB-P1F-15..25	UIX-04
Review & sign-off (P1F)	FCB-P1F-26..30	UIX-02..04
Part numbering	FCB-P2-03..05	PDM-03
TechDraw title block	FCB-P2-06..08	DRW-01
COTS library	FCB-P2-09/10	ROB-01, MOD-05
Headless hooks	FCB-P2-11..14	AUT-01 (done)
Leonardo assembly & register	FCB-P2-15..17	PDM-07, MOD-04 (STEP import checks)
G1 freeze	FCB-P2-18	PDM-07
Review & gates (P2)	FCB-P2-19..21	definition of done
FC-T01 switch criteria	FCB-P2-22	PDM-08, M6 (open question Q1)
Gap audit	FCB-P2A-01/02	ASM-02
Joints & limits	FCB-P2A-03/04	ASM-03
Exploded views & BOM	FCB-P2A-05/06	ASM-04, DRW-02
Sub-assemblies & interference	FCB-P2A-07/08	ASM-05, ASM-06
Robotics kinematics	FCB-P2A-09/10	ROB-05, ASM-10
Tests, review (assembly)	FCB-P2A-11..14	ASM epics' acceptance
COTS robotics parts	FCB-P2R-01..04	ROB-01
Frame & arm templates	FCB-P2R-05/06	ROB-02
Mass/CG & balance	FCB-P2R-07/08	ROB-03, ROB-04, MOD-11
Tests, review (robotics)	FCB-P2R-09..12	ROB epics' acceptance
Owen intake & adopt decision	FCB-P2E-01..04	ECD-01
Board STEP & outline	FCB-P2E-05/06	ECD-02, ECD-03
Placement & enclosure fit	FCB-P2E-07/08	ECD-04
Tests, review (ECAD)	FCB-P2E-09..12	ECD epics' acceptance
MCP base decision	FCB-P2AI-01/02	AUT-03
bruceworx.api & MCP server	FCB-P2AI-03..08	AUT-02, AUT-03
Assistant panel	FCB-P2AI-09..12	AUT-04
Review & try-out (AI)	FCB-P2AI-13..15	AUT-04
OCCT coupling survey	FCB-P3-01..06	KRN-01
STEP harness	FCB-P3-07..11	KRN-02, FND-05
Kernel decision	FCB-P3-12..14	KRN-03
FC-T01 system-of-record lock	decision	superseded by full parity; M6
FC-T02 Mac install & pixi build	task	FND-01
FC-T04 Part-numbering macro	task	PDM-03
FC-T05 TechDraw template	task	DRW-01
FC-T07 COTS library	task	ROB-01
FC-T10 Agent and script hooks	task	AUT-01, AUT-02
Rebase onto 26.3?	decision	answered: track upstream main (FND-03)
Ship P2 as add-on or in fork?	decision	FND-03 (still open; follows the upstream rule "add-ons over core edits")

New epics with no Ops origin (gaps found in this review): FND-06, MOD-01–04, MOD-06–10, MOD-12, ASM-01, ASM-07–09, DRW-03–08, PDM-01, PDM-02, PDM-04–06, all of SIM, MFG-01–04, AUT-05, UIX-05, UIX-06 and UIX-09.

C Reproducing this document

```
# 1. Epics -> roadmap-epics.csv, chapters/gen-epics.tex, chapters/gen-swimlanes.tex
python3 package/bruceworx/docs/fusion-ui-plan/scripts/epics.py

# 2. 3D figures (headless FreeCAD; prints the FACT lines quoted in the text)
build/release/bin/FreeCADCmd package/bruceworx/docs/fusion-ui-plan/scripts/
    render_patterns.py
build/release/bin/FreeCADCmd package/bruceworx/docs/fusion-ui-plan/scripts/
    render_roadmap.py

# 3. UI mocks (FreeCADCmd's Qt has the offscreen plugin)
QT_QPA_PLATFORM=offscreen build/release/bin/FreeCADCmd \
    package/bruceworx/docs/fusion-ui-plan/scripts/run_layout.py

# 4. PDF
cd package/bruceworx/docs/fusion-ui-plan && latexmk -pdf plan.tex
```

Importing into Ops. `roadmap-epics.csv` has one row per epic with columns `id`, `track`, `title`, `milestone`, `milestone_name`, `status`, `depends_on` (space-separated IDs), `origin_ideas`, `summary` and `acceptance`. Statuses reflect the repository (merged PRs), not the Ops sample data.

Sources for external facts (accessed 18–19 Oct 2026).

- conda-forge package metadata via `pixi search` and `https://api.anaconda.org` (OCP, build123d, CadQuery pins).
- SheetMetal add-on: `https://github.com/shaise/FreeCAD_SheetMetal` (LGPL since v0.4.00).
- FreeCAD 1.1 release notes: `https://wiki.freecad.org/Release_notes_1.1`.
- Fusion: `https://www.autodesk.com/products/fusion-360/blog/category/whatsnew`; BOM and change management `https://www.autodesk.com/products/fusion-360/blog/boms-change-management-autodesk-fusion/`; Simulation extension `https://www.autodesk.com/products/fusion-360/simulation-extension.html`.
- SolidWorks 2026: `https://www.javelin-tech.com/3d/drawings-whats-new-in-solidworks-2026/`; `https://blog.cadsoftwaredirect.com/whats-new-in-solidworks-2026/`; PDM editions `https://www.goengineer.com/solidworks-pdm-standard-vs-professional-guide`; Simulation tiers `https://www.javelin-tech.com/blog/2019/07/solidworks-simulation-package-comparison/`.
- Open-source PDM survey: `https://github.com/source-code-learning/FreePDM`; `https://linuxfr.org/users/yboy360/journaux/petit-plm-pour-freecad`.
- build123d / OCP ecosystem: `https://build123d.readthedocs.io/en/latest/external.html`; `https://ocp-freecad-cam.readthedocs.io/`.
- AutoCAD aliases: `https://help.autodesk.com/cloudhelp/2025/ENU/AutoCAD-Customization/files/GUID-D4CACED6-DFBA-43C3-BC42-8D980AB3AE75.htm`.